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MedEvo: Medical Reasoning Evolution through Knowledge Navigation

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Abstract

Medical multimodal reasoning models are typically trained on **teacher-generated rationales**, discarding the valid reasoning in students' own failed trajectories, while general-domain correction methods offer little guidance for errors in image reading, anatomical context, or differential diagnosis. We propose **MedEvo**, which uses **medical knowledge**, namely a tag taxonomy and error-type diagnosis, to navigate both **trajectory repair and data selection**. For each failed rollout, a **gold-blind auditor** locates the first consequential error and provides a minimal, answer-free hint; the same student **retains the valid prefix and regenerates the erroneous suffix**. After each round, MedEvo identifies capability gaps across medical tags on a **held-out feedback set** and prioritizes related questions for new rollouts and repairs. With **supervised fine-tuning (SFT) alone**, MedEvo raises Qwen3.5-9B mean accuracy across seven benchmarks from **65.57% to 69.12%** (47.40% to **54.50%** on the multimodal track of MedXpertQA (MedXpertQA-MM)), surpassing teacher-distillation SFT (66.93%). It also raises Qwen3.5-35B and Qwen3.5-397B mean accuracy from 67.91% to **73.04%** and from 71.83% to **74.95%**, respectively. Our analyses show that failed trajectories contain reasoning worth preserving, that **repair success does not imply learning value**, and that **independent feedback is key to sustained improvement**.

Project: https://pork-pay.github.io/MedEvo/

GitHub: https://github.com/pork-pay/MedEvo

1 Introduction

Medical multimodal reasoning requires connecting visual findings with anatomical context, disease knowledge, and clinical decisions, and expert-level questions remain challenging for current medical vision-language models [2, 3, 4]. Despite recognizing several findings, a model can fail through image misreading, overlooked clinical clues, or unsupported differential diagnosis. Such **failures are often local**: the reasoning before the mistake can remain valid, so preserving valid steps and recovering from the point of failure directly improves medical reasoning [5, 6]. Existing approaches typically use supervised fine-tuning (SFT) to train students on teacher-generated rationales or on answer-correct student rollouts [7, 8, 9]. Discarding a failed trajectory loses its valid reasoning, while replacing it with a complete teacher solution gives the student a path it did not produce. Methods that learn from failures, such as mistake–correction pairs [6], trajectory fusion [10], and Hindsight Hint Distillation (HHD) [1], were developed for mathematics and software engineering, and offer little guidance when an error depends on medical images, anatomy, or differential diagnosis. Moreover, a model's failures reveal which medical capabilities need further training, yet this signal is rarely used to guide the next round of supervision. We therefore ask two questions: how can we repair reasoning while **preserving valid steps**, and how can **failures guide subsequent learning**?

We introduce MedEvo through the principle of **Knowledge Navigation**: using structured medical knowledge and failure diagnosis to guide **where to intervene, how to recover, and what to learn next**. These questions motivate

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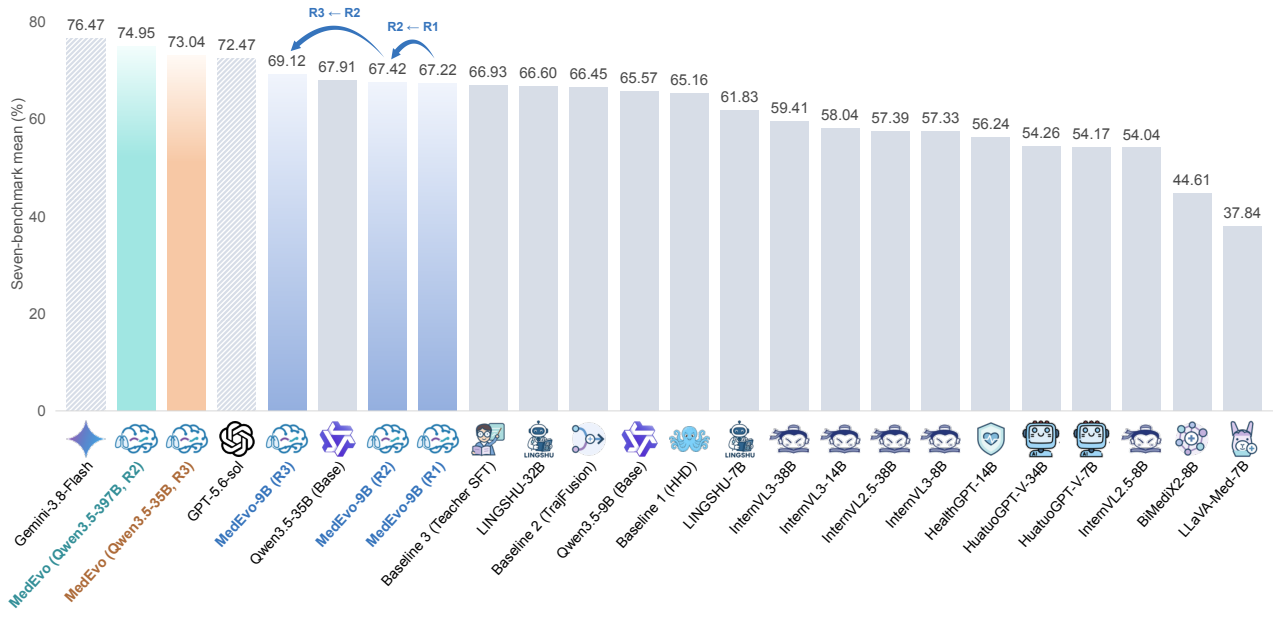


Fig. 1 Medical multimodal reasoning across model scales. Mean accuracy on seven benchmarks. MedEvo improves 9B/35B/397B Qwen3.5 models. B: billion parameters; Hindsight Hint Distillation (HHD) [1].

local recovery decisions and learning decisions reflecting the student's remaining weaknesses. Medical tags locate a failure in its clinical context, while error types identify the reasoning operation that needs correction. We call this navigation because these signals direct a path from an observed failure to a targeted intervention, at two connected levels: **within a trajectory** and **across training rounds** (Fig. 4). For a failed rollout, a gold-blind auditor identifies the first consequential error, preserves the valid prefix, and provides a **minimal, answer-free hint**; the same student then regenerates the erroneous suffix, so the correction remains grounded in its own reasoning (Fig. 2). After each round, MedEvo measures performance and errors across medical tags on a **held-out feedback set** and prioritizes questions for fresh rollouts and repairs, so that recurring failures shape subsequent training.

Across seven medical multimodal benchmarks (Fig. 1), MedEvo improves the mean accuracy of Qwen3.5-9B, Qwen3.5-35B, and Qwen3.5-397B from 65.57%, 67.91%, and 71.83% to **69.12%, 73.04%, and 74.95%**, respectively. With Qwen3.5-9B, it outperforms teacher-distillation SFT (69.12% vs. 66.93%); with Qwen3.5-35B and Qwen3.5-397B, it surpasses GPT-5.6-sol (72.47%); Gemini-3.8-Flash has the highest mean accuracy in this comparison (76.47%).

Our contributions are threefold:

- **Medically informed local repair.** Within trajectories, Knowledge Navigation uses a medical tag taxonomy and error-type diagnosis to locate the first consequential error and a safe rollback boundary; a gold-blind auditor provides an answer-free hint for same-student suffix regeneration that preserves the valid prefix.
- **Feedback-driven data evolution.** Across rounds, Knowledge Navigation uses feedback: capability gaps across medical tags and error types on a held-out feedback set target questions and failures for rollout and repair.
- **Empirical gains and findings.** With SFT alone, MedEvo improves Qwen3.5-9B, Qwen3.5-35B, and Qwen3.5-397B across seven benchmarks. Our analyses show that failed trajectories contain reasoning worth preserving, that repair success does not imply learning value, and that independent feedback is key to sustained improvement.

2 Related Work

2.1 Medical Multimodal Reasoning and Supervision Quality

Medical vision-language models use biomedical instruction tuning [2, 3], generalist training in Lingshu [11], and bilingual BiMediX2 training [12]. MedGemma builds medical vision-language models on Gemma 3 [13]. Structured knowledge can also be integrated with multimodal learning through knowledge graphs [14]. Beyond answer accuracy, **process-level audits** separate visual recognition, knowledge application, and evidence integration [15, 16], and rubric-based evaluations can miss hallucinations, motivating **factuality checks** [17].

Supervision quality depends on **reasoning content and student compatibility**: process supervision provides step-level feedback [18], on-policy distillation studies model compatibility [9], and ChainPrune [19] reduces redundant rea-

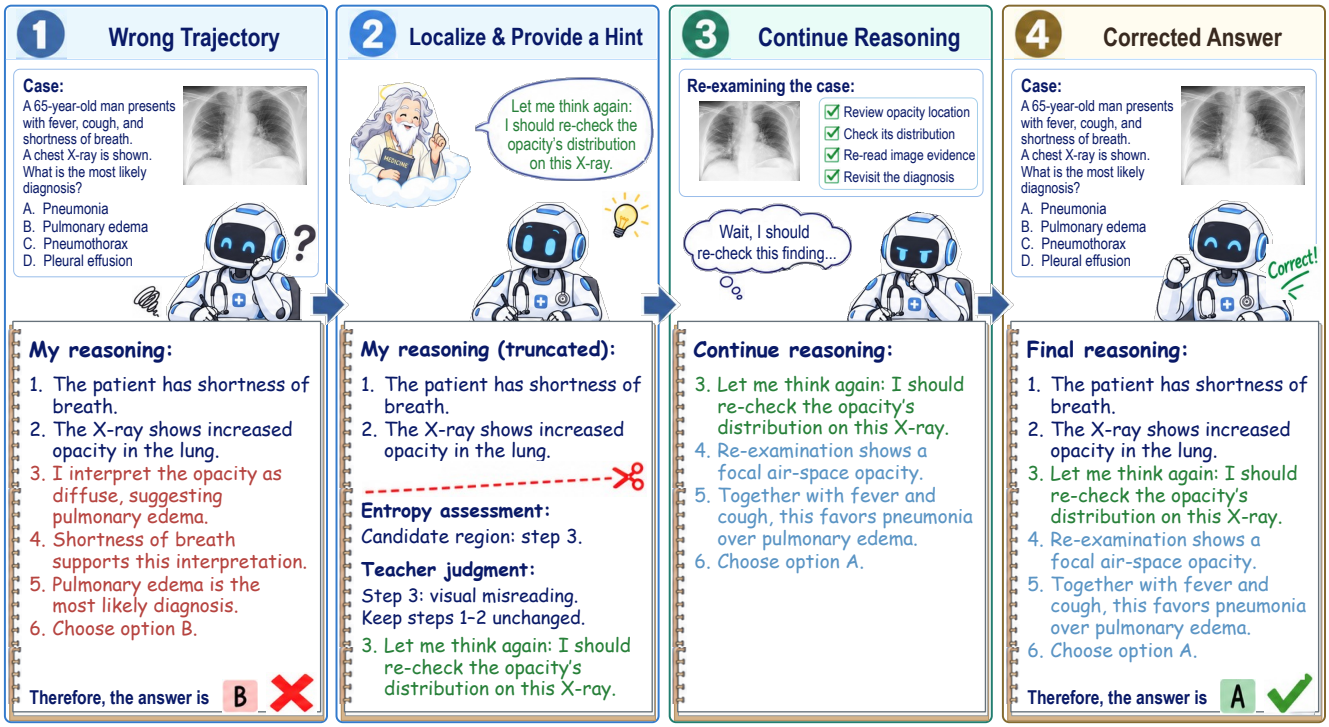


Fig. 2 Local trajectory repair. A minimal, answer-blind hint guides the student to recover from a failed reasoning step while retaining its supported prefix. Localization and admission checks are detailed in § 3.2.1 and § 3.2.4.

soning while distinguishing structural efficiency from token length. JOLT makes student compatibility explicit: even a higher-performing privileged teacher can induce an unhelpful student update, motivating joint teacher–student training with regularization toward the student's current policy [20]. MedEvo diagnoses failures with a medical codebook, screening repairs for answer correctness, valid boundaries, integrity, and student-relative learnability.

2.2 Learning from Failures and Localized Repair

Failed trajectories can supply supervision: Learning from Mistakes (LEMA) trains on mistake–correction pairs [6], TrajFusion combines incorrect and correct trajectories [10], and HHD guides recovery with concise hints [1]. Identifying an error can enable its correction [5], and backtracking distillation and selective regeneration study where correction should begin [21, 22]. MedEvo differs in three respects: the corrected suffix is **generated by the same student** rather than written by a teacher; the auditor **never accesses the reference answer**; and both localization and hints are conditioned on medical tags and error types. Self-Retrospection Distillation uses completed interactions, including failed attempts, to supervise pre-interaction predictions of required knowledge and likely pitfalls in the same policy [23]. MedEvo uses the repaired trajectory as its training target: it retains the supported prefix and learns from the student's regenerated continuation.

2.3 Data Evolution and Recursive Self-Improvement

Self-Taught Reasoner (STaR) and Reinforced Self-Training with Expectation–Maximization (ReST^{EM}) build supervision by filtering self-generated reasoning for correctness [7, 8], and micro-reflective self-distillation adds capability-dependent trajectory selection [24]. MediSkill-Evo [25] instead evolves validated clinical knowledge banks without updating the backbone. Work on frozen medical models similarly evolves external procedures, supported facts, and visual references, checking candidate updates on both new and previously seen cases while leaving model parameters unchanged [26]. MedEvo follows STaR and ReST^{EM} self-training but differs in two respects: rather than STaR's rationalization, which supplies the correct answer as a hint [7], it repairs failures with answer-free hints; and it selects each round's data using **medical-tag and error-type feedback on a fixed feedback set**, linking local recovery to what the student should learn next. Knowledge Navigation couples repair and feedback selection. Appendix F reviews medical hallucinations, reasoning supervision, correction, and self-evolution.

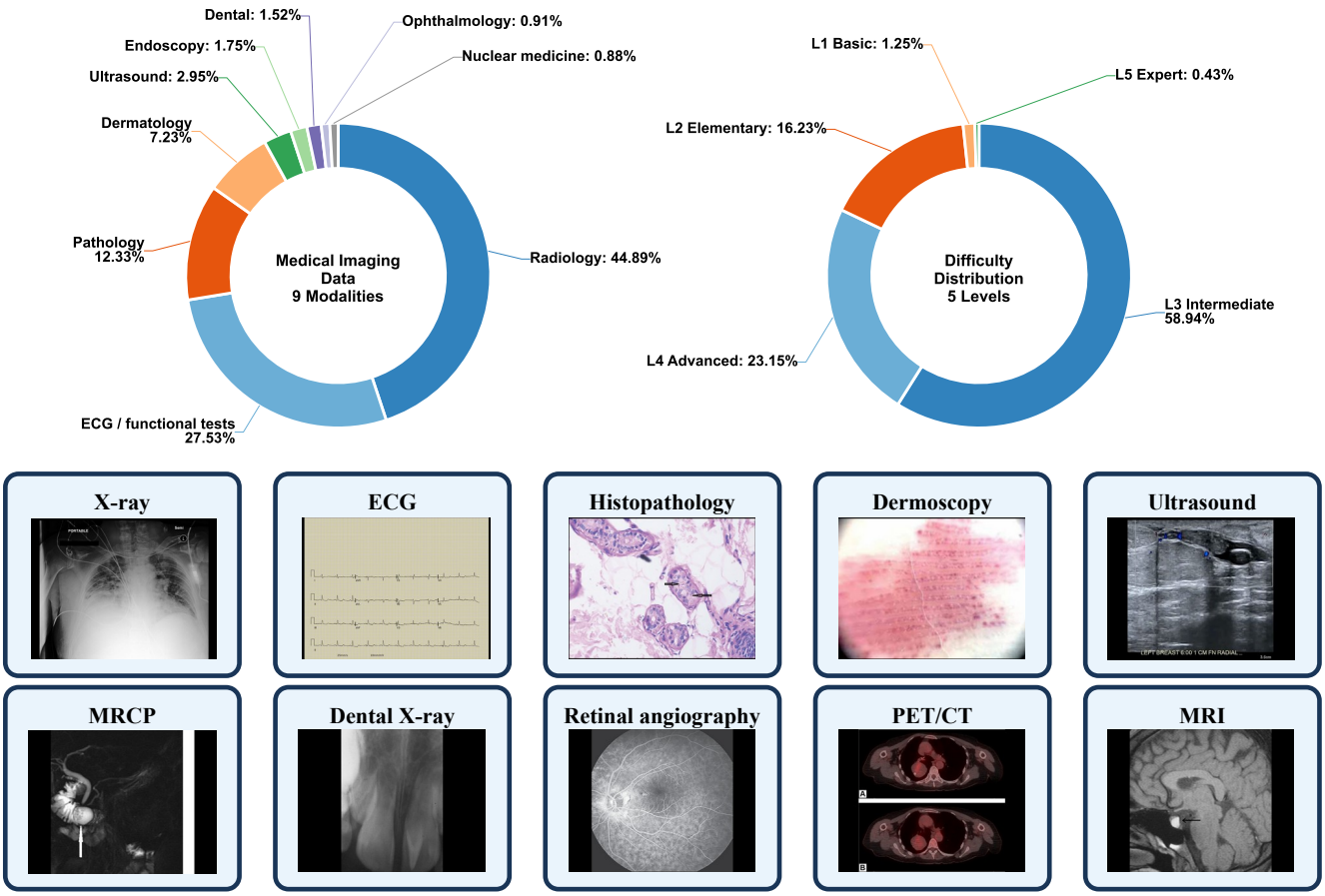


Fig. 3 Medical multimodal data. Top: modality and difficulty; bottom: examples. Electrocardiogram (ECG); magnetic resonance cholangiopancreatography (MRCP); positron emission tomography/computed tomography (PET/CT); magnetic resonance imaging (MRI). Examples include both anatomical images and physiological recordings.

3 Method

Knowledge Navigation. We define **Knowledge Navigation** as a feedback-guided process that uses structured domain knowledge and failure diagnosis to choose **where to intervene, how to recover, and what to learn next**. MedEvo instantiates this process with a medical tag taxonomy and an error codebook: tags organize the clinical context, error types identify the failed reasoning operation, and their associated recovery operations guide intervention. Within a trajectory, the process selects a safe rollback boundary and an answer-free hint while preserving supported reasoning (§ 3.2). Across rounds, measured capability gaps guide question selection and repair priorities (§ 3.3). The term navigation therefore describes the link from a structured diagnosis to an actionable decision and its subsequent reassessment. The taxonomy supplies the reference structure; student failures guide learning (Fig. 4).

Learning protocol. A question $x_i = (I_i, q_i, O_i)$ consists of one or more medical images I_i, the clinical context and question q_i, and the answer options O_i. Following STaR and ReST^{EM} [7, 8], each round t uses the current student $M_t = \pi_{\theta_t}$ to generate trajectories, collects correct rollouts and accepted repairs into the training set D_{t+1}, and fine-tunes the next student from the original model M_0 rather than from M_t. The training pool, feedback set, and test benchmarks are kept separate throughout.

3.1 Medical Taxonomy and Failure Diagnosis

3.1.1 Medical Question Tags

We assign each question a four-dimensional hierarchical tag

$$z_i = (z_i^{\mathrm{mod}}, z_i^{\mathrm{anat}}, z_i^{\mathrm{dis}}, z_i^{\mathrm{diff}}), \tag{1}$$

covering modality, anatomy, disease, and difficulty. Tags serve two roles. During repair, the auditor receives only the **observable subset** z_i^{obs}, i.e., tags supported by the input itself, excluding answer-derived labels such as the

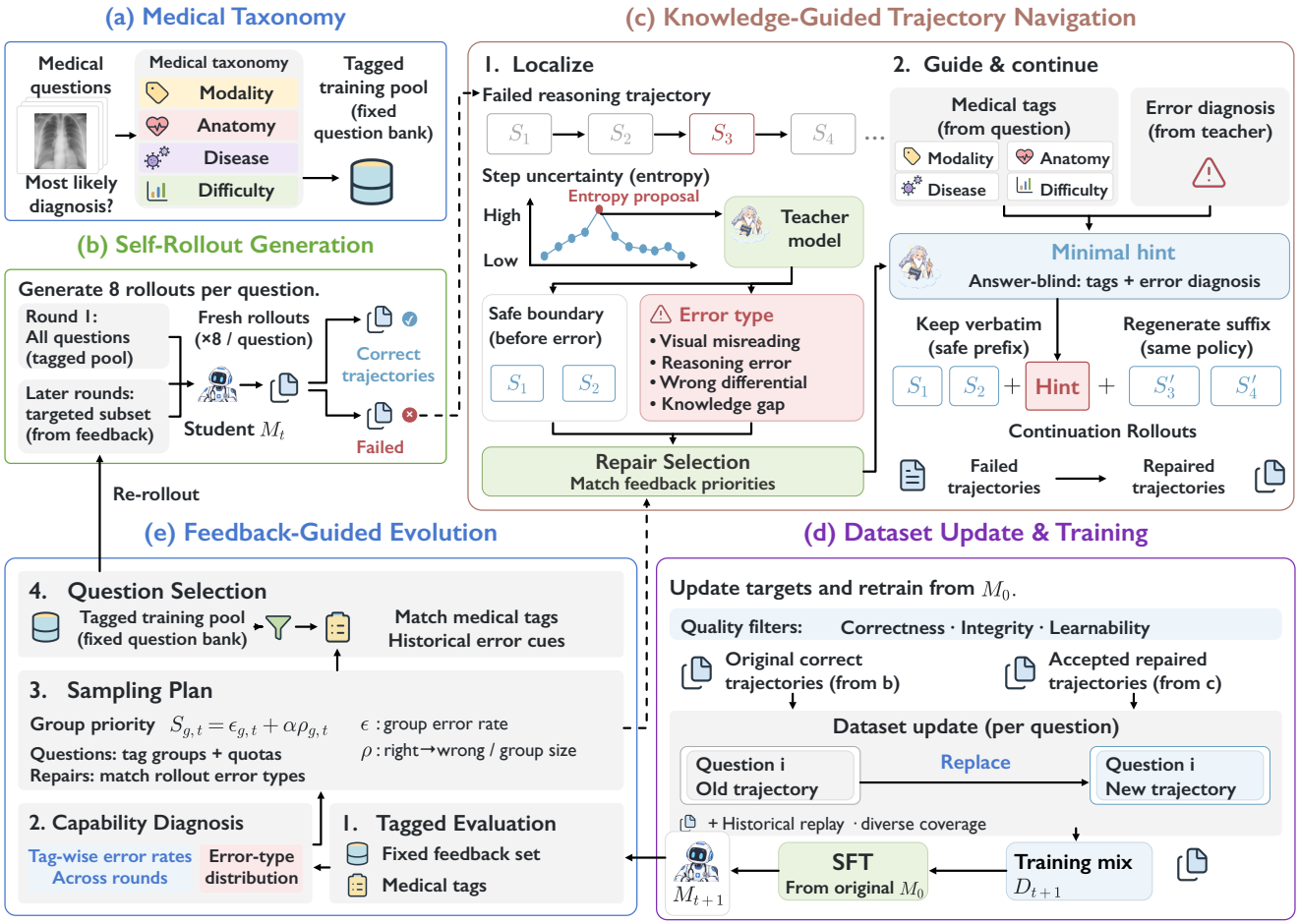


Fig. 4 Knowledge Navigation in MedEvo. (a) Medical tags organize questions. (b) The current student generates fresh rollouts. (c) Knowledge-guided trajectory navigation localizes errors and provides recovery hints through a gold-blind auditor; the student preserves the valid prefix and regenerates the suffix. (d) Admitted trajectories update the training set, retaining valid historical targets; the next model trains from M_0. (e) Fixed-set feedback guides question selection for the next round.

diagnosed disease. Across rounds, tag-level accuracy on the feedback set identifies capability gaps and guides question selection. See Appendix C.1 for the tag hierarchy and annotation rules.

3.1.2 Error Types and Their Relation to Repair

We diagnose failures at the process level, building on work that separates medical reasoning from knowledge and audits intermediate steps [15, 16]. Our codebook defines **four parent error types** with five subtypes each, twenty in total (Appendix C.1.2). Each type is paired with a recovery operation used for hint construction (§ 3.2.2): visual misreading calls for perceptual re-examination; reasoning error for reconstructing the evidence-to-decision link; differential diagnosis error for comparing discriminative features; and knowledge gap for a narrowly scoped knowledge cue. Whereas tags describe the question, error types describe what went wrong in a specific trajectory; because diagnoses vary across rollouts and checkpoints, **each fresh failure is diagnosed independently**.

3.2 Knowledge-Guided Trajectory Navigation

3.2.1 Student Rollouts and Uncertainty-Based Localization

Tyen et al. [5] show that large language models can correct reasoning errors once the error location is given; we therefore **localize an error before regenerating**. For each selected question, M_t samples eight trajectories, and answer-correct ones proceed to quality screening (§ 3.2.4). An incorrect or unparsable trajectory is segmented into semantically complete steps

$$r^- = (s_1, \ldots, s_K), \tag{2}$$

such as image observation, feature interpretation, mechanism, differential diagnosis, and option mapping. The goal is to keep the steps before the **first consequential error** and regenerate the rest.

To propose where to audit, we compute the student's token entropy

$$H_j = -\sum_{v \in \mathcal{V}} \pi_{\theta_t}(v \mid x, r^-_{<j}) \log \pi_{\theta_t}(v \mid x, r^-_{<j}), \tag{3}$$

where $\mathcal{V}$ is the vocabulary, j indexes tokens, and $r^-_{<j}$ is the token prefix. Logs are natural, with $0 \log 0 = 0$. For each nonempty step s_k, let $\mathcal{J}_k$ be its token-index set and $H_{s_k} = (H_j)_{j \in \mathcal{J}_k}$. We combine their **mean and 90th percentile** $Q_{0.9}$, and measure the increase over the preceding $m_k = \min(m, k-1)$ steps:

$$E^*_k = \lambda \frac{1}{|\mathcal{J}_k|} \sum_{j \in \mathcal{J}_k} H_j + (1-\lambda)\, Q_{0.9}(H_{s_k}), \tag{4}$$

$$\Delta E_k = \begin{cases} 0, & k = 1, \\ E^*_k - \dfrac{1}{m_k} \displaystyle\sum_{\ell=k-m_k}^{k-1} E^*_\ell, & 2 \le k \le K, \end{cases} \tag{5}$$

For $k \ge 2$, $m_k > 0$. After min–max normalizing both profiles (constant profiles map to zero), we select the anchor

$$k^* \in \arg\max_{1 \le k \le K} \left[\widetilde{E^*_k} + \widetilde{\Delta E_k} \right], \tag{6}$$

and the audit window covers steps $k^* \pm 1$, within valid indices, using $\lambda = 0.5$ and $m = 3$ (Appendix A.2). **Entropy proposes the audit region**; the auditor locates errors and boundaries using images, questions, and reasoning (§ 3.2.2).

3.2.2 Gold-Blind Auditing and Medical Error Diagnosis

Building on hint-guided recovery in HHD [1], an external auditor $\mathcal{A}$ (GPT-5.6-sol) turns the diagnosis into concise guidance. The auditor is gold-blind: it receives the question and images, the failed trajectory, the entropy proposal $\mathcal{E}_t(x, r^-)$, and the observable tags z^{obs}, but **no reference answer, reference rationale, or answer-derived metadata**. It returns

$$(a, e, b, h) = \mathcal{A}\big(x, r^-, \mathcal{E}_t(x, r^-), z^{\text{obs}}\big), \tag{7}$$

where $a \in \{\textsc{Repair}, \textsc{Skip}\}$ is the action. For $\textsc{Repair}$, e is the **earliest consequential error** together with its diagnosed type, b is the last fully valid step before e, and h is a **minimal recovery hint**. For $\textsc{Skip}$, (e, b, h) are unused and no continuation is generated. The hint instantiates the recovery operation of the diagnosed type (§ 3.1.2) in the observable context. For example, a perceptual hint may ask the student to reinspect a region or waveform attribute without stating the finding, and a differential hint may request comparison of a discriminative feature without naming the diagnosis. Hints must not reveal the answer, an answer option, or a corrected rationale. The audit prompt appears in Appendix D.3.

3.2.3 Prefix Preservation and Same-Student Continuation

Like LEMA [6], we train on corrected mistakes, but the correction is generated by the student itself. The boundary splits the failed trajectory into a **preserved prefix and an erroneous suffix**,

$$r^- = p_b \oplus s^-, \qquad p_b = (s_1, \ldots, s_b). \tag{8}$$

Here $b \in \{0, \ldots, K-1\}$, p_0 is empty, and $\oplus$ denotes token-sequence concatenation. **We keep p_b verbatim**, append the hint rendered as a natural-language recovery cue $g(h)$, and let the same checkpoint continue:

$$s^+ \sim \pi_{\theta_t}\big(\cdot \mid x, p_b \oplus g(h)\big), \qquad r^* = p_b \oplus g(h) \oplus s^+. \tag{9}$$

The question stays in the user message, $p_b \oplus g(h)$ prefills an unfinished assistant message, and the structured audit is stored only as metadata. With the prefix fixed, we sample up to five suffixes per failed trajectory, **stopping at the first admitted continuation**, and send answer-correct continuations to admission (§ 3.2.4; Appendix D.4).

3.2.4 Repair Admission and Training Targets

A continuation is admitted only if it passes **answer, boundary, integrity, and learnability checks**. Learnability is scored on the **new suffix alone**, so that a valid prefix cannot mask the difficulty of the correction. Adapting the instruction-following difficulty of Li et al. [27], we measure how much the suffix depends on the task input:

$$D_{\text{repair}} = \frac{\mathcal{L}_t(s^+ \mid x, p_b)}{\mathcal{L}_t(s^+ \mid p_b)}, \tag{10}$$

For a token sequence s and context c, $\mathcal{L}_t(s \mid c) = -\sum_{j=1}^{|s|} \log \pi_{\theta_t}(s_j \mid c, s_{<j})$, in nats. Both terms are **computed without the hint**; the denominator omits x but retains p_b. A smaller ratio indicates a larger relative loss reduction from explicitly conditioning on x beyond the prefix, which may itself contain task information. Both suffixes must be nonempty, all losses finite, and the denominator strictly positive. We also measure how strongly the student still prefers its original wrong suffix, using per-token losses to account for length differences:

$$\Delta_{\text{repair}} = \frac{\mathcal{L}_t(s^+ \mid x, p_b)}{|s^+|} - \frac{\mathcal{L}_t(s^- \mid x, p_b)}{|s^-|}. \tag{11}$$

For a repair candidate $\xi = (x, r^-, r^*, b)$, the gap is measured in nats per token. A positive value means the student assigns lower average loss to the wrong suffix than to the repair. The learnability gate L_ξ requires $D_{\text{repair}} \in [0.35, 0.85]$ and $\Delta_{\text{repair}} \in [0.02, 0.30]$, with **thresholds calibrated on the feedback set** and frozen before final evaluation (Appendix A.4). Admission is

$$\text{Accept}(\xi) = C_\xi \wedge A_\xi \wedge N_\xi \wedge L_\xi, \tag{12}$$

Here C_ξ checks correctness, A_ξ validates the **prefix and boundary**, and N_ξ rejects unsupported additions or repetition. Each admitted repair is trained as the complete response $r^* = p_b \oplus g(h) \oplus s^+$, including the recovery cue, with **loss on all assistant tokens**:

$$\mathcal{L}_{\text{repair}}(\theta) = -\sum_{j=1}^{|r^*|} \log \pi_\theta(r^*_j \mid x, r^*_{<j}). \tag{13}$$

Correct rollouts share this objective; image and user tokens are excluded from the loss. If no repair is admitted, we retain a qualified correct rollout or a valid historical target; if neither is available, we add none. **Evaluation excludes hints and audit outputs.**

3.3 Feedback-Driven Evolution of Supervision

Across-round Knowledge Navigation follows the iterative self-training pattern of STaR and ReST^{EM} [7, 8], but uses **medical-tag and error-type feedback on a fixed set** to allocate generation and repair effort. The cycle is assessment, targeted selection, local repair, screened dataset update, and retraining. The taxonomy and error codebook remain fixed; **selection priorities and the supervision dataset evolve** as the student changes.

3.3.1 First-Round Coverage

The training pool $\mathcal{T}$ and a separate, fixed feedback set V_{feedback} are drawn from public medical datasets (§ 4.1). In the first round, M_0 generates **rollouts for every question** in $\mathcal{T}$; correct trajectories enter screening, failures are repaired as in § 3.2, and accepted targets form D_1, giving $M_1 = \text{SFT}(M_0, D_1)$.

3.3.2 Feedback across Medical Tags and Error Types

Before constructing D_{t+1}, we evaluate M_t on the feedback set, including M_0 at initialization. For a supported tag group g with $n_g > 0$, the error rate is

$$\epsilon_{g,t} = \frac{1}{n_g} \sum_{i \in V_{\text{feedback},g}} (1 - c_{i,t}), \qquad n_g = |V_{\text{feedback},g}|, \tag{14}$$

where $c_{i,t} \in \{0, 1\}$ indicates correctness. Comparing consecutive checkpoints separates corrections, regressions, persistent errors, and retained successes (Appendix A.5), and diagnosing failed responses yields each group's error-type distribution. Groups below a support threshold back off to their parent tag. Feedback questions guide selection and threshold calibration but **never enter training**, and test results never influence data construction.

3.3.3 Selecting Questions and Updating the Dataset

From the second round on, we allocate the generation budget across groups by the rule-based priority

$$S_{g,t} = \epsilon_{g,t} + \alpha\, \rho_{g,t}, \qquad \rho_{g,t} = \frac{N^{10}_{g,t}}{n_g}, \tag{15}$$

where $t \geq 1$ and $N^{10}_{g,t}$ counts correct-to-incorrect transitions from M_{t-1} to M_t and $\alpha \geq 0$ weights regressions. Quotas are proportional to $S_{g,t}$, capped by the available unique questions, and rounded deterministically; unused capacity is redistributed, and each multi-tag question is counted once (Appendix A.5). Within each group, candidates are

Algorithm 1: MedEvo Training with Local Repair and Feedback-Guided Data Evolution

Require: Initial student M_0; training pool $\mathcal{T}$ with verification labels; fixed feedback set V_{feedback}; gold-blind auditor $\mathcal{A}$; rounds R; selection and generation budgets.

$\mathcal{G} \leftarrow \text{AssignMedicalTags}(\mathcal{T}, V_{\text{feedback}})$
$D_0 \leftarrow \varnothing$; $H \leftarrow \varnothing$ *// Active targets and trajectory history*
for $t \leftarrow 0$ **to** $R-1$ **do**
  // Feedback assessment and question selection
  $F_t \leftarrow \text{Evaluate}(M_t, V_{\text{feedback}}, \mathcal{G})$
  if $t = 0$ **then**
    $Q_t \leftarrow \mathcal{T}$ *// Broad first-round coverage*
  else
    $P_t \leftarrow \text{PlanQuotas}(F_t, F_{t-1}, \mathcal{G})$
    $Q_t \leftarrow \text{RetrieveUnique}(\mathcal{T}, P_t, H)$
  $D_{t+1} \leftarrow D_t$
  // Fresh student rollouts and local trajectory repair
  foreach $x \in Q_t$ **do**
    $\mathcal{R}_x \leftarrow \text{Rollout}(M_t, x, 8)$
    $C_x \leftarrow \text{ScreenCorrect}(x, \mathcal{R}_x)$
    foreach failed $r^- \in \text{PrioritizeFailures}(\mathcal{R}_x, F_t)$ within budget **do**
      $u \leftarrow \text{StepUncertainty}(M_t, x, r^-)$
      $(a, e, b, h) \leftarrow \mathcal{A}(x, r^-, u, z^{\text{obs}}(x))$ *// Observable tags only*
      $H \leftarrow \text{RecordHistory}(H, x, r^-, a, e, b, h)$
      if $a = \textsc{Skip}$ **then**
        continue
      $p_b \leftarrow \text{ValidPrefix}(r^-, b)$
      for $j \leftarrow 1$ **to** 5 **do**
        $s^+ \sim M_t(\cdot \mid x, p_b \oplus g(h))$
        $r^* \leftarrow p_b \oplus g(h) \oplus s^+$
        $\xi \leftarrow (x, r^-, r^*, b)$
        if Accept(ξ) (Eq. 12) **then**
          $C_x \leftarrow C_x \cup \{r^*\}$
          break
    // Keep one active target per question; retain valid history on ties
    $D_{t+1} \leftarrow \text{RetainOrReplace}(D_{t+1}, x, C_x)$
  // Full-response SFT from the same initial student
  $M_{t+1} \leftarrow \text{SFT}(M_0, D_{t+1})$ *// Reset optimizer and schedule*
return $\{M_t, D_t\}_{t=1}^{R}$

retrieved by tag match and ranked by historical error records. Selected questions receive **fresh rollouts** from M_t, their failures are diagnosed anew, and failures whose error types match the feedback priorities are repaired first. Feedback thus decides where effort is spent, while the checks in § 3.2.4 decide what enters training.

The dataset keeps **at most one active target per question**. An admitted candidate replaces the existing target only if a blinded comparison judges it better in evidence support, reasoning consistency, and conciseness; otherwise the existing target is kept, and all versions are logged separately. Writing each target as a pair (x, r) keyed by question ID, with $D_0 = \varnothing$, replaced targets $R_t \subseteq D_t$, accepted replacements U_{t+1}, and new targets A_{t+1} for previously unrepresented questions,

$$D_{t+1} = (D_t \setminus R_t) \cup U_{t+1} \cup A_{t+1}, \tag{16}$$

$$M_{t+1} = \text{SFT}(M_0, D_{t+1}). \tag{17}$$

U_{t+1} contains exactly one replacement for each question in R_t; A_{t+1} contains one target per new question, disjoint by question ID from D_t. Each round restarts from M_0 with a fresh optimizer and learning-rate schedule, so **experience is carried only through the evolving dataset**. Algorithm 1 summarizes the procedure.

4 Experiments

4.1 Experimental Setup

Models and data. Our main backbone is Qwen3.5-9B [28]; Qwen3.5-35B and Qwen3.5-397B test scalability (§ 5.3). The training pool contains 81,293 questions from the training splits of public medical datasets, spanning nine modality categories and five difficulty levels (Fig. 3), and is deduplicated by question and source identifiers, patient/case identifiers when available, image hashes, and near-duplicate clusters (Appendix C.1.4). A separate, fixed set of 5,000 modality-stratified questions serves as the feedback set and **never enters training**. Dataset roles and training configurations are given in Appendices B.1 and C.1.5.

Tab. 1 Repair workload across rounds. Rate: fraction repaired; arrows compare rounds.

Training round	R1	R2	R3
Generator	R0	R1	R2
Sent for repair	**27,233**	**25,831**↓	**23,620**↓
Share of question bank	33.50%	31.78%	29.06%
Successfully repaired	19,567	18,258	16,797
Repair success rate	**71.85%**	**70.68%**↓	**71.11%**↑
Mean repair attempts	3.37	2.48	2.12

Evaluation. We use MedUniEval [29], the Hulu-Med implementation built on Lingshu's MedEvalKit [11], for medical visual question answering (VQA) on PMC-VQA [30], VQA-RAD [31], SLAKE [32], PathVQA [33], OmniMedVQA [34], the multimodal track of MedXpertQA (MedXpertQA-MM) [4], and the medical subset of Massive Multi-discipline Multimodal Understanding (MMMU), denoted MMMU-Med [35]. All models share the **answering prompt and parser** (Appendix D.1). We report accuracy (%) and the unweighted seven-benchmark mean; accuracy gains and Δ denote absolute differences in percentage points.

Tab. 2 Medical reasoning benchmarks. General-purpose and medical model accuracy (%). MedEvo: R3. Avg.: unweighted seven-benchmark mean; MedXQA: MedXpertQA-MM; MMMU-Med: the medical subset of MMMU. 🧰: medical models; 🔧: trajectory-repair methods. Blue: MedEvo. Bold/underline: top two per group and benchmark, with shared ranks for ties.

	Model	PMC-VQA	VQA-RAD	SLAKE	PathVQA	OmniMed VQA	MedXQA	MMMU-Med	Avg.
	Proprietary models								
	GPT-5.6-sol	60.43	70.29	**81.34**	54.84	76.67	77.50	86.24	72.47
	Gemini-3.8-Flash	**64.52**	**74.72**	**81.34**	**58.58**	**80.76**	**86.30**	**89.10**	**76.47**
	Open-source models (< 10B)								
🧰	LLaVA-Med-7B [2]	30.50	53.70	48.00	38.80	44.30	20.30	29.30	37.84
🧰	HuatuoGPT-V-7B [36]	53.30	67.00	67.80	48.00	74.20	21.60	47.30	54.17
🧰	BiMediX2-8B [12]	43.50	49.20	57.70	37.00	63.30	21.80	39.80	44.61
	InternVL2.5-8B [37]	51.30	59.40	69.00	42.10	81.30	21.70	53.50	54.04
	InternVL3-8B [38]	53.80	65.40	72.80	48.60	79.10	22.40	59.20	57.33
🧰	LINGSHU-7B [11]	56.30	67.90	**83.10**	**61.90**	82.90	26.70	54.00	61.83
	Qwen3.5-9B (Base) [28]	59.20	68.10	76.80	48.80	82.20	47.40	76.50	65.57
🔧	Baseline 1 (HHD) [1]	58.93	67.65	71.45	50.12	82.35	48.75	76.90	65.16
🔧	Baseline 2 (TrajFusion) [10]	59.32	69.12	77.23	49.23	83.46	49.56	77.20	66.45
	Baseline 3 (Teacher SFT)	61.23	70.15	71.60	53.20	81.90	52.30	78.14	66.93
🧰🔧	**MedEvo (9B)**	**62.68**	**70.40**	78.23	53.34	**85.41**	**54.50**	**79.28**	**69.12**
	Open-source models (> 10B)								
🧰	HealthGPT-14B [39]	56.40	65.00	66.10	56.70	75.20	24.70	49.60	56.24
🧰	HuatuoGPT-V-34B [36]	56.60	61.40	69.50	44.40	74.00	22.10	51.80	54.26
	InternVL3-14B [38]	54.10	66.30	72.80	48.00	78.90	23.10	63.10	58.04
	InternVL2.5-38B [37]	57.20	61.40	70.30	46.90	79.90	24.40	61.60	57.39
	InternVL3-38B [38]	56.60	65.40	72.70	51.00	79.80	25.20	65.20	59.41
🧰	LINGSHU-32B [11]	57.90	**76.50**	**89.20**	**65.90**	83.40	30.90	62.30	66.59
	Qwen3.5-35B (Base) [28]	62.40	70.50	70.10	49.15	83.20	60.00	80.00	67.91
🧰🔧	**MedEvo (35B)**	**64.63**	75.83	80.30	56.01	**83.97**	**66.50**	**84.02**	**73.04**

Baselines. On Qwen3.5-9B, we compare with two methods that learn from failed trajectories, HHD [1] and TrajFusion [10], and with **teacher-distillation SFT** on rationales generated by Qwen-3.8-Max. External general-

purpose and medical models provide additional context (Appendix G).

4.2 Main Results

Tab. 2 reports the results. On Qwen3.5-9B, MedEvo raises mean accuracy from 65.57% to 69.12% and **improves all seven benchmarks**, with the largest gain on MedXpertQA-MM [4] (47.40% → 54.50%). It outperforms HHD [1], TrajFusion [10], and teacher-distillation SFT, whose mean accuracies are 65.16%, 66.45%, and 66.93%, respectively. On Qwen3.5-35B, mean accuracy rises from 67.91% to 73.04% (§ 5.3). A question-level paired bootstrap (10,000 resamples; Tab. 3, right) yields **95% confidence intervals (CIs) that exclude zero** on every benchmark. These intervals capture question-level uncertainty; variation across training runs is reported in Tab. 8.

Tab. 3 Evolution and statistical uncertainty (Qwen3.5-9B). Left: accuracy (%) across rounds, each trained from Base (R0). Right: absolute gain R3−R0 and 95% bootstrap confidence intervals (CIs) (10,000 paired question resamples). The mean CI resamples each benchmark independently before averaging. All gain CIs exclude zero. Arrows compare consecutive rounds; MedXQA annotations give absolute changes. Bold/underline: best/second-best per row.

	Round-wise accuracy				Paired bootstrap	
Benchmark	R0	R1	R2	R3	Δ (R3–R0)	95% CI
PMC-VQA	59.20	59.54 ↑	<u>60.97</u> ↑	**62.68** ↑	+3.48	[+1.9, +5.1]
VQA-RAD	68.10	<u>68.47</u> ↑	68.06 ↓	**70.40** ↑	+2.30	[+0.8, +3.8]
SLAKE	76.80	76.92 ↑	<u>77.34</u> ↑	**78.23** ↑	+1.43	[+0.2, +2.7]
PathVQA	48.80	<u>52.38</u> ↑	51.27 ↓	**53.34** ↑	+4.54	[+2.6, +6.5]
OmniMedVQA	<u>82.20</u>	80.81 ↓	81.58 ↑	**85.41** ↑	+3.21	[+1.5, +4.9]
MedXQA	47.40	52.56 ↑ 5.16	<u>52.79</u> ↑ 0.23	**54.50** ↑ 1.71	+7.10	[+5.2, +9.0]
MMMU-Med	76.50	<u>79.83</u> ↑	**79.91** ↑	79.28 ↓	+2.78	[+1.3, +4.3]
Avg.	65.57	67.22 ↑	<u>67.42</u> ↑	**69.12** ↑	+3.55	[+2.4, +4.7]

4.3 Accuracy Gains and Repair Workload

Mean accuracy increases from 65.57% to 67.22%, 67.42%, and 69.12% across rounds (Tab. 3, left), while the number of questions sent for repair falls from 27,233 to 23,620 and repair success remains near 71% (Tab. 1). Thus, the later rounds improve benchmark performance with a smaller question-level repair workload; § 5.2 isolates the role of feedback-guided selection. Individual benchmarks still fluctuate, as illustrated by PathVQA [33]. Modality and error-type statistics on the fixed feedback set guide subsequent selection (Appendix B.3); the **seven held-out benchmarks assess generalization**.

5 Ablation Studies and Analysis

We analyze repair, feedback, scale, and **learning value**, using Qwen3.5-9B except in scaling experiments [28].

5.1 Repair Design in the First Round

Tab. 4 Repair design (Qwen3.5-9B, R1). Repair success and downstream accuracy (%); Base accuracy is 65.57%. Type-guided suffix continuation performs best on both. Red ↓: below the full method; bold: best.

Method	Repair success	Mean accuracy
Hint + full rewrite	57.80↓	63.20↓
No-hint rewrite	36.70↓	61.16↓
Generic hint + suffix continuation	64.50↓	64.20↓
Type-guided hint + suffix continuation (Ours)	**71.80**	**67.22**

Tab. 4 compares four first-round designs varying regeneration scope (full rewrite vs. suffix continuation) and hint specificity (none, generic, or type-guided). We measure repair success and downstream mean accuracy. **Type-guided suffix continuation performs best** with 71.80% repair success and 67.22% mean accuracy. Generic-hint continuation achieves 64.50% repair success and 64.20% accuracy; hint-guided full rewriting achieves 57.80% and 63.20%. Both prefix preservation and error-type conditioning contribute.

5.2 Contribution of Feedback across Rounds

Tab. 5 isolates the contribution of feedback. Sharing the R1 data-generation model and historical dataset (R1 accuracy: 67.22%), four variants run two more rounds: repair without feedback, feedback without medical tags, feedback without error types, and full MedEvo. Full MedEvo reaches 69.12% at R3 versus 66.67%, 67.53%, and 66.21% for the three variants, respectively. Without feedback or without error-type feedback, accuracy falls below the shared R1 reference; without medical tags, it improves only marginally (67.22% to 67.53%). **Combined medical-tag and error-type feedback performs best**.

Tab. 5 Feedback ablation (Qwen3.5-9B). Mean accuracy (%) for branches sharing the R1 generator and historical dataset (R1: 67.22%); each round of SFT restarts from M_0. Full MedEvo achieves the highest R3 accuracy. Arrows compare with R1; Δ = R3 − R1.

Method	R2	R3	Δ
No feedback (repair only)	66.39↓	66.67↓	−0.55
Without medical tags	67.32↑	67.53↑	+0.31
Without error-type feedback	67.01↓	66.21↓	−1.01
Full MedEvo	**67.42**↑	**69.12**↑	**+1.90**

5.3 Generalization across Model Scales

We apply MedEvo to Qwen3.5-35B for three rounds and to Qwen3.5-397B for two rounds, evaluating on the same seven benchmarks [4, 30, 31, 32, 33, 34, 35] (Tab. 6). On Qwen3.5-35B, mean accuracy rises from 67.91% to 69.73%, 70.81%, and 73.04%. MedXpertQA-MM [4] peaks at 66.73% in R2 and declines slightly to 66.50% in R3, indicating benchmark-dependent optima. On Qwen3.5-397B, mean accuracy rises from 71.83% to 74.71% and 74.95%, and MedXpertQA-MM from 67.65% to 75.00%. The R2 model improves six of seven benchmarks, with OmniMedVQA [34] essentially unchanged (85.10% → 85.05%). At both scales, **mean accuracy increases in every round**.

Tab. 6 Scaling to larger models. Accuracy (%) for Qwen3.5-35B (left) and Qwen3.5-397B (right). Every round trains from Base (R0). Avg.: seven-benchmark mean. Arrows compare consecutive rounds; MedXQA annotations give absolute changes. Bold/underline: best/second-best within each model scale and benchmark.

	35B				397B		
Benchmark	R0	R1	R2	R3	R0	R1	R2
PMC-VQA	62.40	63.12↑	60.33↓	**64.63**↑	63.47	64.87↑	**65.72**↑
VQA-RAD	70.50	62.08↓	70.91↑	**75.83**↑	74.72	76.05↑	**77.15**↑
SLAKE	70.10	78.14↑	79.18↑	**80.30**↑	77.38	78.89↑	**79.10**↑
PathVQA	49.15	52.71↑	50.87↓	**56.01**↑	52.49	**58.51**↑	57.90↓
OmniMedVQA	83.20	**85.39**↑	85.22↓	83.97↓	85.10	**85.15**↑	85.05↓
MedXQA	60.00	65.68↑ +5.68	**66.73**↑ +1.05	66.50↓ −0.23	67.65	74.40↑ +6.75	**75.00**↑ +0.60
MMMU-Med	80.00	81.02↑	82.41↑	**84.02**↑	82.03	**85.10**↑	84.75↓
Avg.	67.91	69.73↑	70.81↑	**73.04**↑	71.83	74.71↑	**74.95**↑

5.4 Learning Value of Repair Supervision

Beyond repair success, we examine three questions about **what makes repair supervision useful** (Fig. 5).

Which trajectories are worth repairing? We compare the 8,000 questions with the longest failed trajectories against 8,000 random questions, each used in one round. The longest set has lower repair success (55.70% vs. 67.50%) but yields higher downstream accuracy (66.78% vs. 64.35%; Fig. 5a), favoring **learning-based trajectory selection**.

How many hints help? All main experiments provide one hint per failed trajectory. Using the longest-8k subset from the previous comparison, we vary the number of hints per failed trajectory from 1 to 3, 5, and 8, keeping five continuation attempts. Mean accuracy is 66.78% with one hint and 67.01% with three, then falls to 66.74% and 64.10% with five and eight (Fig. 5b). Three hints yield the highest observed accuracy, 0.23 percentage points above one hint; additional hints reduce accuracy.

Which feedback guides evolution? Training-error feedback raises training accuracy from 63.23% to 78.00% but lowers benchmark accuracy from 65.57% to 64.01%; fixed-set feedback yields 71.00% training and 69.12% benchmark accuracy (Fig. 5c,d). Failures supply repairs; **independent feedback reveals generalizable weaknesses**.

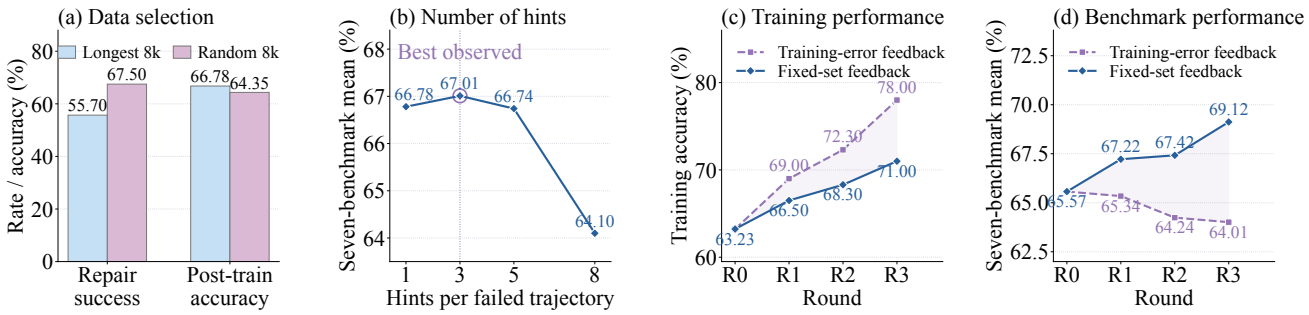


Fig. 5 Selection, hints, and feedback (Qwen3.5-9B). (a) Longest-8k versus random-8k questions: lower repair success can yield higher downstream accuracy. (b) Hint-count ablation uses the same longest-8k subset and run configuration as (a), with five continuation attempts; $k = 1$ reproduces its 66.78% single-hint baseline. Three hints perform best. (c,d) Training accuracy and seven-benchmark mean with training-error or fixed-set feedback. Feedback: 5,000 held-out questions.

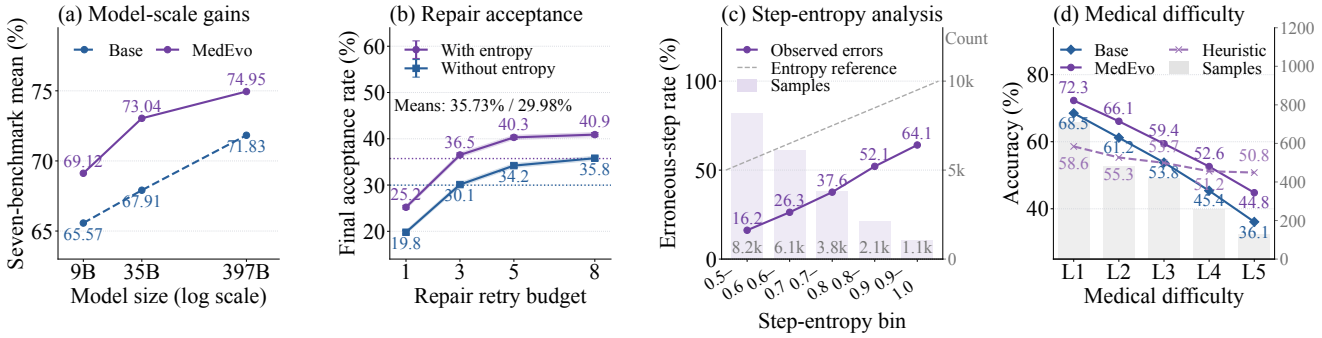


Fig. 6 Scale, uncertainty, and difficulty. (a) Mean accuracy after R3 for 9B/35B and R2 for 397B. (b) Repair acceptance with/without entropy localization; pointwise 95% Wilson intervals use 45,000 attempts per setting. (c) Error rates and step counts by min–max-normalized step-entropy bin; the diagonal is $y = 100x$. (d) Accuracy on a modality-stratified 2,000-question subset of the feedback set; counts use the right axis. Heuristic is a training-free HHD-inspired hint-recovery baseline (Appendix A.6).

Further analyses. Fig. 6 summarizes additional repair analyses. Entropy localization increases final acceptance, defined as the fraction of repair candidates retained, at every continuation budget: from 19.8–35.8% to 25.2–40.9% for one to eight attempts (Fig. 6b). The erroneous-step rate rises from 16.2% to 64.1% across increasing entropy bins, supporting **entropy as an audit prior** (Fig. 6c). Relative to Base, MedEvo improves accuracy from 68.5% to 72.3% at L1 and from 36.1% to 44.8% at L5 (Fig. 6d). MedEvo exceeds Heuristic at L1–L4, while Heuristic attains 50.8% at L5 versus 44.8% for MedEvo.

6 Reliability Analysis

We assess **repair reliability** through human audit and **MedEvo stability** across independent runs.

6.1 Human Audit

Two reviewers independently assess each trajectory using the same rubric and resolve disagreements through discussion. Pass rates use the resulting adjudicated binary labels. Cohen's κ uses the independent labels before adjudication. Tab. 7 reports results for the audited sample of repaired trajectories. Annotators identify 95.5% of submitted trajectories as erroneous ($\kappa = 0.90$) and validate 84.5% of the selected repair boundaries ($\kappa = 0.85$). Hints are answer-blind in 97.2% of cases ($\kappa = 0.94$). Repaired answers are judged correct in 88.3% of cases ($\kappa = 0.89$), and 78.5% of repairs are judged **image-grounded**. Across the four error families, answer-correctness pass rates range from 88% to 89%, and grounding pass rates from 76% to 82%.

The audit assesses **complementary answer correctness and visual grounding**, with marginal pass rates of 88.3% and 78.5%, respectively. Agreement is higher for answer correctness ($\kappa = 0.89$) than for grounding ($\kappa = 0.70$). These findings support assessing answers and the visual evidence behind reasoning.

Tab. 7 Human audit of repaired trajectories. Adjudicated binary pass rates (%) and pre-adjudication Cohen's κ. The 95% bootstrap CIs are for κ, resampling trajectories with the two independent ratings paired. V/R/D/K: visual, reasoning, differential-diagnosis, and knowledge errors. Repaired-output checks are shaded blue.

		Agreement		Error family			
Audit criterion	Pass (%)	κ	95% CI	V	R	D	K
Original trajectory erroneous	95.5	0.90	[0.83, 0.97]	96	96	94	96
Valid repair boundary	84.5	0.85	[0.66, 0.88]	88	84	82	84
No answer leakage in hint	97.2	0.94	[0.87, 1.00]	96	100	96	97
Image-grounded repaired reasoning	**78.5**	0.70	[0.58, 0.82]	76	80	76	82
Correct repaired answer	**88.3**	0.89	[0.81, 0.97]	88	89	88	88

6.2 Stability across Random Seeds

Tab. 8 reports three independent training runs from the same Base checkpoint. The **improvement is highly reproducible**. Every run follows the ordering Base < R1 < R2 < R3, and R3 averages 69.09% ± 0.24, compared with 65.57% for Base. Even the weakest run reaches 68.84%, so the improvement far exceeds the variation across seeds. All three runs improve more from R2 to R3 than from R1 to R2.

Each round retrains from M_0, so **the evolving dataset carries information across rounds**. Longer training runs can characterize the timing and saturation of these gains.

Tab. 8 Stability across training runs (Qwen3.5-9B). Seven-benchmark mean accuracy (%), three runs; all improve through R3. Sample standard deviation (SD); Δ: absolute gain over Base. Bold/underline: top two rounds; blue: R3.

Model	Seed 1	Seed 2	Seed 3	Mean ± SD	Δ
Base	65.57	65.57	65.57	65.57 ± 0.00	—
R1	67.22	67.05	67.38	67.22 ± 0.17	↑ 1.65
R2	<u>67.42</u>	<u>67.21</u>	<u>67.60</u>	<u>67.41</u> ± 0.20	↑ 1.84
R3	**69.12**	**68.84**	**69.31**	**69.09 ± 0.24**	↑ 3.52

7 Conclusion and Discussion

We introduced MedEvo to study how failed medical reasoning trajectories can become useful supervision. **Knowledge Navigation** links local repair to subsequent data selection: a gold-blind auditor provides a minimal, answer-free hint, the student preserves supported reasoning and regenerates the erroneous suffix, and a fixed feedback set directs later repairs toward remaining weaknesses. **With SFT alone**, MedEvo improves Qwen3.5-9B, Qwen3.5-35B, and Qwen3.5-397B across seven benchmarks, with consistent gains across the three Qwen3.5-9B runs.

The central lesson from these experiments is that **correcting a response and constructing useful supervision are different objectives**. Preserving valid reasoning, selecting repairs for learning value, and using independent feedback each affect what the student learns. Their contribution should therefore be assessed through downstream learning, alongside repair success and evidence grounding.

7.1 Insights for Medical Self-Evolution

The practical question is how to allocate repair effort: which parts of a failed trajectory remain useful, which corrections should become training targets, and which capability gaps warrant new data.

From diagnosis to intervention. Medical tags describe the clinical context, while error types identify a recovery operation for the current trajectory. The two guide different decisions: tag-level feedback prioritizes questions, and local error diagnosis guides the hint and rollback boundary. The repair and feedback ablations (Tabs. 4 and 5) support this division of roles within MedEvo.

Evolution through supervision. The taxonomy and error codebook remain fixed in the present implementation; **selection priorities and accepted supervision change with the student**. Each round retrains from M_0 on the updated dataset, so information passes across rounds through supervision. Regressions and persistent weaknesses then redirect subsequent repair effort.

Failed trajectories retain valuable reasoning. Suffix continuation outperforms full rewriting in both repair success and downstream accuracy (Tab. 4), supporting the value of **preserving the valid prefix**. In the trajectory-selection experiment, the longest-8k subset has lower repair success than random selection (55.70% vs. 67.50%) but yields higher downstream accuracy (66.78% vs. 64.35%; Fig. 5a). Repair success alone would therefore rank these two choices differently from their downstream learning outcomes.

Repairability and learning value are different objectives. The comparison above shows a divergence between repair success and learning value in the tested subsets; it does not isolate trajectory length from question difficulty. In the hint-count experiment, three hints yield the highest observed accuracy, only 0.23 percentage points above one hint, while five and eight hints perform worse (Fig. 5b). **Guidance should be calibrated against subsequent learning**, rather than increased solely to ease correction. Localization has a related distinction: the erroneous-step rate rises from 16.2% to 64.1% across entropy bins (Fig. 6c), making entropy a useful inspection cue in this analysis, while the evidence-grounded audit determines whether an error is present and where regeneration should begin.

The source of feedback matters. Training-error feedback improves fit to the training data while benchmark accuracy declines; fixed-set feedback improves benchmark accuracy across rounds (Fig. 5c,d). In these experiments, **training failures supply repair candidates, while independent feedback helps prioritize capability gaps**. The distinction prevents progress on the training examples from serving as the sole criterion for allocating later repair effort.

Answer correctness and grounding are complementary. The human audit assesses answer correctness and visual grounding separately (§ 6.1). Its results motivate checking **evidence use alongside accuracy** and hint non-leakage when admitting repaired trajectories as supervision.

7.2 Limitations and Future Directions

The current implementation has a fixed knowledge structure. Its taxonomy and error codebook constrain which weaknesses can be represented, and a strong proprietary auditor supplies the diagnoses. Evaluating open-source auditors and separating the effects of trajectory length from question difficulty would clarify these dependencies. Our benchmarks use multiple-choice and short-answer formats; deployment requires **clinical evaluation**.

From diagnosis to grounded recovery. Future work could study **evidence-grounded recovery** through source-attributed retrieval, factuality checks, and image-region supervision, evaluating answer leakage, grounding, and downstream learning jointly.

Adaptive navigation. Persistent failures could guide expert-validated updates to the taxonomy and recovery operations. Longer runs on independent data would help assess the regression, cost, and saturation of such **knowledge-structure evolution**.

Transferable learning strategies. Further studies could test repair transfer across modalities, specialties, and institutions, while accounting for annotation and auditor budgets. Relevant outcomes include transfer, rare-condition coverage, and robustness under distribution shift.

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Reproducibility Statement. The main text and appendices describe the method, prompts, training configurations, and evaluation procedures. The present release is limited to the manuscript; the implementation and training-data files are not released. Use of the underlying public datasets remains subject to their licenses and access restrictions.

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A Method Details

This appendix provides implementation details that complement §3. Within round t, **the generator $M_t = \pi_{\theta_t}$ is frozen**: it produces rollouts, entropy profiles, continuations, and learnability scores, while the next student is trained from M_0 (Eq. 17).

A.1 Failure Definition and Step Segmentation

Given a question $x = (I, q, O)$ with reference answer $y \in O$, M_t samples a trajectory r and a parsed answer $\hat{y}$. A trajectory r^- fails if $\hat{y} \neq y$ or no valid answer can be parsed. Each failed trajectory is segmented into K **semantically complete steps** (Eq. 2), typically covering image observation, feature interpretation, mechanism, differential diagnosis, and option mapping. A boundary $b \in \{0, \ldots, K-1\}$ splits r^- into the preserved prefix $p_b = (s_1, \ldots, s_b)$ and the erroneous suffix $s^- = (s_{b+1}, \ldots, s_K)$ (Eq. 8), where $b = 0$ denotes an empty prefix.

A.2 Entropy-Based Localization

Step scores. Token entropy H_j, the step score E_k^*, and the local surge ΔE_k follow Eqs. 3–5, with entropy measured in nats. We use $\lambda = 0.5$ and $m = 3$, with $m_k = \min(m, k-1)$ for $k > 1$ and $\Delta E_1 = 0$. Combining the mean with the 90th percentile $Q_{0.9}$ balances average uncertainty with an upper-tail signal within each step.

Anchor and window. Both profiles are **min–max normalized** within each trajectory, giving $\widetilde{E}_k^*$ and $\widetilde{\Delta E}_k$; a constant profile maps to zero. The anchor is $k^* \in \arg\max_{1 \le k \le K}(\widetilde{E}_k^* + \widetilde{\Delta E}_k)$, and the audit window is

$$\mathcal{W}(k^*) = \{s_{\max(1,\, k^* - w_\ell)}, \ldots, s_{\min(K,\, k^* + w_r)}\}, \qquad w_\ell = w_r = 1. \tag{18}$$

The step preceding the window defines a provisional boundary

$$b_0 = \max(0,\ k^* - w_\ell - 1). \tag{19}$$

Both $\mathcal{W}(k^*)$ and b_0 are proposals; **the auditor sets the final boundary** b (Appendix A.3).

Proposal assessment. For every audited trajectory, the auditor classifies the entropy proposal as HIT (the first consequential error lies at or adjacent to k^*), PARTIAL_HIT (the window covers the propagated error, but the unsupported premise begins earlier), MISS (the first error lies outside the window), or NON_ERROR_UNCERTAINTY (the high-entropy region is medically and logically valid). These logged labels support Fig. 6c.

A.3 Gold-Blind Audit and Recovery Operators

Auditor. We use GPT-5.6-sol as the auditor $\mathcal{A}$ in Eq. 7. Its inputs are the images, the question and options, the failed trajectory, the entropy profile (E_k^*, normalized scores, ΔE_k, k^*, and $\mathcal{W}(k^*)$), and **observable tags with provenance**. The reference answer, reference rationale, answer-derived tags, and retrieval metadata are **excluded through allowlist serialization** (Appendix C.1.3). The auditor returns a structured record (Appendix D.3) containing an action in {Repair, Skip}, the first consequential error e with its type and subtype, the boundary b with a verbatim ending span (from s_b if $b > 0$, empty if $b = 0$), and the hint h. It returns Skip when only answer formatting or option selection is wrong and no reasoning revision is justified. The final boundary may differ from b_0. The auditor **traces evidence dependencies backward** and moves the boundary earlier when a later entropy surge stems from an upstream unsupported premise; conversely, it retains a high-entropy step that is medically and logically valid.

Recovery operators. An error type selects a recovery operator o; the text hint h instantiates that operator in natural language. The operator belongs to

$$o \in \{\mathcal{P}_{\text{inspect}},\ \mathcal{R}_{\text{reconstruct}},\ \mathcal{D}_{\text{disambiguate}},\ \mathcal{K}_{\text{bridge}}\}, \tag{20}$$

each paired with a parent error type (§3.1.2); subtype-level operations are listed in Appendix C.1.2. Operators define recovery actions.

- **Perceptual re-examination** ($\mathcal{P}_{\text{inspect}}$) names a region or attribute to revisit: lesion distribution, boundary shape, waveform morphology, stain localization, or cross-view consistency, without stating the finding.
- **Inference reconstruction** ($\mathcal{R}_{\text{reconstruct}}$) asks students to link evidence, mechanism, diagnosis, and decision, for example when supportive and causal treatments are confused or care steps are misordered.
- **Differential disambiguation** ($\mathcal{D}_{\text{disambiguate}}$) requests comparison of discriminative features: contrast pattern, onset profile, lesion location, signature pathology, or an exclusion criterion, without naming the diagnosis.
- **Knowledge bridging** ($\mathcal{K}_{\text{bridge}}$) supplies a narrowly scoped concept, such as a diagnostic criterion, drug mechanism, or immunohistochemical localization rule. It may state the concept but must not map it to an answer option.

Rendering. The cue $g(h)$ consists of a natural transition followed by the hint, with flexible wording. Error labels and audit explanations are **stored as metadata** and are never shown to the student.

A.4 Continuation, Admission, and Training Targets

Continuation budget. The current checkpoint generates eight rollouts per selected question. Each failed trajectory selected for repair is audited once and receives a single hint. With $p_b \oplus g(h)$ fixed, up to five suffixes are sampled (Eq. 9); **sampling stops at the first admitted continuation**. The user message contains x, the unfinished assistant message contains $p_b \oplus g(h)$, and generation continues within the same message **without repeating or modifying the prefix** (Appendix D.4).

Learnability gate. Using the suffix scores D_{repair} and Δ_{repair} (Eqs. 10–11), computed under M_t **without the hint**,

$$L_\xi = \mathbb{1}\left[D_{\min} \le D_{\text{repair}}(\xi) \le D_{\max} \ \wedge \ \Delta_{\min} \le \Delta_{\text{repair}}(\xi) \le \Delta_{\max}\right], \tag{21}$$

with $[D_{\min}, D_{\max}] = [0.35, 0.85]$ and $[\Delta_{\min}, \Delta_{\max}] = [0.02, 0.30]$. The lower bound on Δ_{repair} requires that the student still prefers its original wrong suffix, so the repair carries information the student has not yet acquired. The upper bound limits the increase in per-token loss relative to the wrong suffix. Thresholds are selected on the feedback set and **frozen before final evaluation**. Feedback-set questions are **excluded from training targets**. Thresholds, retained fractions, and generator hashes are logged with each repair record (Appendix C.1.4).

Admission checks. Admission follows Eq. 12. C_ξ compares the parsed answer with the reference label. A_ξ **verifies that the exported prefix matches** the auditor's verbatim span and that b precedes the diagnosed error. N_ξ **rejects continuations** that introduce symptoms, history, measurements, or findings absent from the input, as well as repetition loops. Reject candidates with either suffix empty, nonfinite losses, or a nonpositive denominator in Eq. 10.

Training targets and fallback. Admitted repairs are exported as $r^\star = p_b \oplus g(h) \oplus s^+$ and trained with Eq. 13; original correct rollouts and historical repairs use the same **full-response loss**, and image and user tokens are masked. If no repair is admitted for a question, a quality-screened correct rollout or the existing valid target is retained; if neither exists, the question contributes no target. **No hint or auditor output is used at evaluation**.

A.5 Feedback Statistics, Quota Allocation, and Dataset Update

Feedback statistics. Besides the error rate in Eq. 14, consecutive checkpoints yield transition counts

$$N_{g,t}^{uv} = \sum_{i \in V_{\text{feedback},g}} \mathbb{1}[c_{i,t-1} = u \ \wedge \ c_{i,t} = v], \qquad u, v \in \{0,1\}, \quad t \ge 1, \tag{22}$$

which separate **corrected errors** (N^{01}), **regressions** (N^{10}), **persistent errors** (N^{00}), and **retained successes** (N^{11}). The priority in Eq. 15 normalizes regressions by n_g. For diagnostics, we also report the conditional regression rate $N_{g,t}^{10}/(N_{g,t}^{10} + N_{g,t}^{11})$ only when its denominator is positive; it is undefined and omitted when the preceding checkpoint has no correct examples. Groups comprise supported single-axis tags and predeclared intersections. A group with $n_g < n_{\min}$ backs off to its nearest supported ancestor, and identical resolved nodes are merged. Each question is counted once within a group, and overall statistics are computed over the deduplicated union of questions.

Quota allocation. From the second round on, the budget B_t is allocated in proportion to $S_{g,t}$ (Eq. 15), using largest-remainder rounding with deterministic tie-breaking. Quotas are **capped by the available unique training questions**, and surplus capacity is **redistributed among eligible groups**. Each multi-tag question is charged to the budget once. If all scores are zero, questions are sampled uniformly from eligible unique questions. Token and auditor-call caps impose additional limits, and any unfilled quota is recorded. The priority and quota rule is Eq. 15; the planning template in Appendix D.2 assembles a plan and tag-based retrieval requests subject to these numerical constraints. The validator in Appendix C.1.4 checks the plan before execution.

Retrieval and repair priority. Within each group, candidates are retrieved by tag match and ranked by historical error records. Candidates may include questions already used in training. Selected questions receive **fresh rollouts** from M_t, and their failures are diagnosed anew. Failures matching both the modality and the error type prioritized by feedback are repaired first, followed by modality-only or error-type-only matches. Matching uses **current diagnoses**. Priority determines where repair effort is spent; admission (Appendix A.4) determines what enters training.

Dataset update. Questions are deduplicated by canonical ID, and each keeps **at most one active target**. An admitted candidate replaces the existing target only if a comparison auditor, blinded to round and model labels, judges it better in evidence support, reasoning consistency, and conciseness; **ties retain the existing target**. All versions and rollout-specific diagnoses are logged in a separate history. Each round retrains from M_0 on the complete updated dataset (Eqs. 16–17) with a fresh optimizer and learning-rate schedule. Generation caps are fixed in advance, and plan records follow Appendix C.1.4.

A.6 Settings of Analysis Experiments

Unless stated otherwise, analyses use Qwen3.5-9B with the training configuration in Tab. 14 and evaluation in § 4.1.

Repair designs (Tab. 4). All variants are applied in the first round under the same rollout and continuation budgets. Hint + full rewrite uses the type-guided hint but regenerates the trajectory from the start of the response. No-hint rewrite regenerates the full trajectory without a hint. Generic hint + suffix continuation keeps the auditor's boundary but replaces the type-guided hint with a fixed generic cue.

Feedback ablations (Tab. 5). All variants share the R1 data-generation model and historical dataset and use the same budgets. R2 data are generated by the shared R1 model; R3 data are generated by each branch's own R2 model. In every branch and round, SFT starts from M_0 with a fresh optimizer and learning-rate schedule. No feedback omits feedback-guided question selection. Without medical tags omits medical-tag feedback. Without error-type feedback uses tag-group quotas but does not match repairs to prioritized error types.

Trajectory selection (Fig. 5a). The longest-8k subset contains the 8,000 questions whose failed trajectories are longest; the random-8k subset is sampled uniformly. Each subset is used for one round of continuation training from M_0.

Hint count (Fig. 5b). Using the same longest-8k subset, M_0 initialization, and one-round training configuration as Fig. 5a, we use $k \in \{1, 3, 5, 8\}$ hints per failure and five continuation attempts. The $k = 1$ setting reproduces the single-hint baseline of 66.78%.

Feedback source (Fig. 5c,d). Training-error feedback selects by training errors, replacing feedback-set errors.

Retry budget and entropy (Fig. 6b). We vary the continuation budget over $\{1, 3, 5, 8\}$ with and without entropy localization. Final acceptance is the admitted fraction over 45,000 attempts per setting, with 95% Wilson intervals.

Step-entropy analysis (Fig. 6c). The horizontal axis uses min–max-normalized step entropy rather than raw entropy in nats. The five equal-width bins have boundaries $0.5, 0.6, 0.7, 0.8, 0.9, 1.0$; the statistical unit is a reasoning step. Each bin reports the erroneous-step rate and the number of steps. The dashed line is the numerical reference $y = 100x$, with y expressed as a percentage, rather than an error-probability calibration target.

Difficulty analysis and Heuristic baseline (Fig. 6d). We sample 2,000 questions from the 5,000-question feedback set using modality-stratified sampling. The difficulty counts are 710, 480, 420, 260, and 130 for L1–L5, following the rubric in Appendix C.1.1. This subset is used only for difficulty analysis and does not enter training; capability-gap assessment in every round uses the full 5,000-question feedback set. Thus, this analysis characterizes performance on feedback-set questions.
Heuristic is an HHD-inspired hint-recovery variant [1], evaluated on the same difficulty-analysis subset as MedEvo. It takes the image and question as input and performs direct inference without additional training. It uses hint-based recovery without error localization, prefix preservation, or feedback-driven data evolution. This variant serves as a difficulty-analysis baseline. At L5, it attains 50.8% accuracy versus 44.8% for MedEvo; MedEvo is higher at L1–L4. The L5 result indicates that HHD-style hint recovery remains competitive on the hardest stratum.

B Dataset Statistics

B.1 Dataset Roles and Evaluation Scope

The study separates the training bank, the fixed feedback set, and the final evaluation benchmarks. Training questions supply supervision; feedback guides diagnosis and selection; benchmarks measure performance.

Tab. 9 Dataset roles and sizes. Test coverage is counted in benchmarks.

Resource	Size	Role
Training bank	81,293 questions	Source of original correct rollouts, accepted repairs, and retained valid targets.
Fixed feedback set	5,000 questions	Medical capability diagnosis and subsequent data selection; excluded from training.
Final evaluation	7 benchmarks	Independent downstream accuracy reporting.

The training bank is constructed from public medical training splits and deduplicated using question and source identifiers, available patient or case identifiers, image hashes, and near-duplicate clusters. See Appendix C.1.4 for structured records and plan validation, and Appendix C.1.3 for information-access rules. Fig. 3 shows nine modality categories and five difficulty levels; the tag hierarchy is described in Appendix C.1.1.
Evaluation uses MedUniEval from Hulu-Med [29], built on Lingshu's MedEvalKit [11]. The seven benchmarks are PMC-VQA [30], VQA-RAD [31], SLAKE [32], PathVQA [33], OmniMedVQA [34], MedXpertQA-MM [4], and the medical subset of MMMU [35]. Their accuracies are averaged without weighting by benchmark size. Modality and error-family results below describe the feedback set, separately from benchmark evaluation.

B.2 Round-wise Data Construction

Tab. 1 highlights a declining repair workload across training rounds. The number of questions submitted for repair falls from **27,233 to 23,620**, a reduction of **3,613 questions (13.27%)** relative to R1. Their share of the fixed question

bank decreases from 33.50% to 29.06%. The unit of this workload measure is a submitted question.
The final training set remains fixed at **81,293 questions** in each round, combining qualified original correct trajectories with **accepted repairs** and retained valid historical targets. Thus, repair workload decreases while training coverage stays constant. The number of successfully repaired questions decreases from 19,567 to 16,797 as fewer questions are submitted. Repair success rates are 71.85%, 70.68%, and 71.11%, remaining stable across rounds. With at most five continuation attempts per repair, the mean number of attempts also falls from **3.37 to 2.12**, indicating improved hint-guided correction efficiency. Each column identifies the generator and training round.

B.3 Additional Experimental Details

B.3.1 Evolution across Twenty Error Subtypes

Fig. 7 shows twenty subtypes across four families. Bars normalize rates by subtype R0 values; labels retain reported percentages. All subtypes are below R0 at R3. The knowledge-gap family has an R3/R0 ratio of approximately 0.89, compared with 0.72–0.75 for the other families, indicating a smaller relative reduction in the reported breakdown, conditional on a consistent counting denominator across rounds. Rates remain separate by subtype.

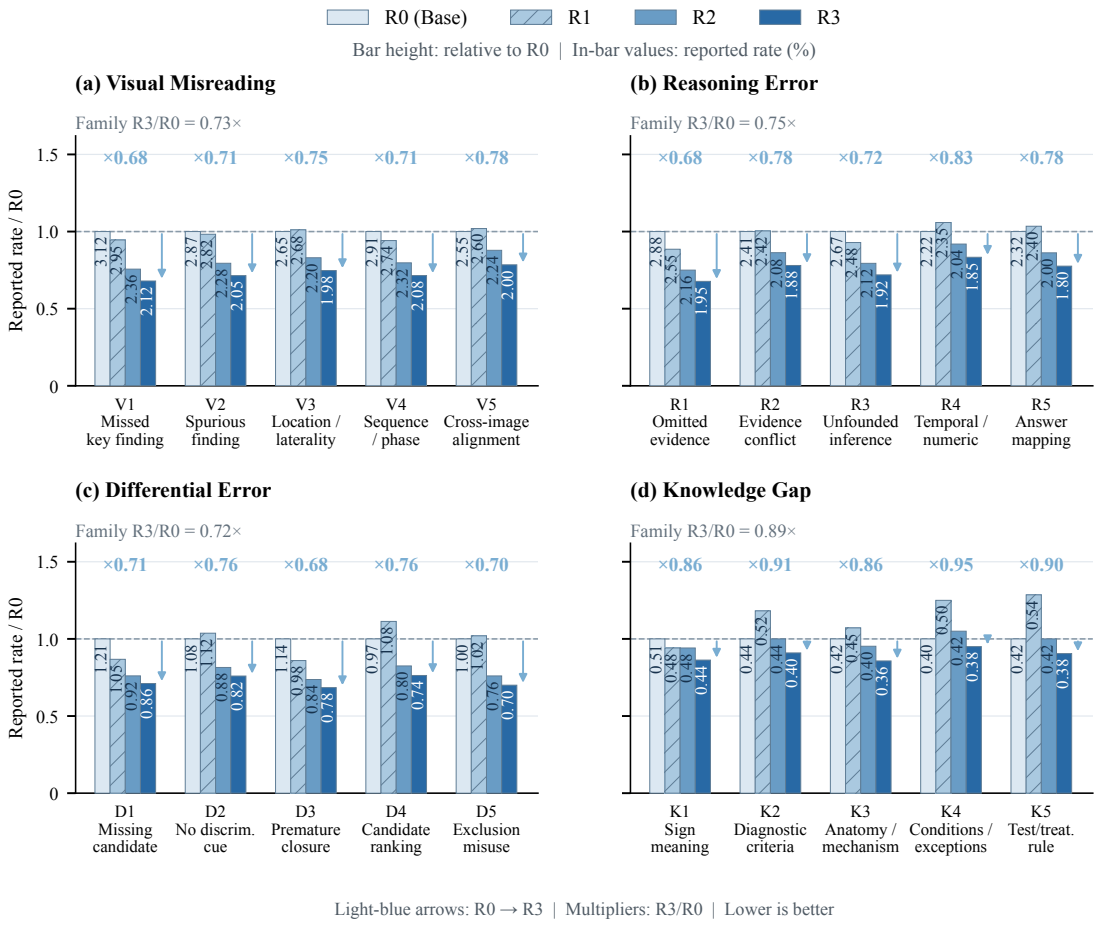


Fig. 7 Error-subtype evolution (Qwen3.5-9B). Bar heights normalize each subtype rate by R0; labels show original percentages. Arrows and multipliers compare R3 with R0. Family ratios divide summed R3 rates by summed R0 rates. Horizontal R1–R5 codes identify reasoning-error subtypes; legend R0–R3 labels identify training rounds.

B.3.2 Round-wise Feedback Statistics

Tab. 10 and 11 report the per-round modality error rates and error-family percentages; § 4.3 summarizes the modality trends. Blue downward arrows indicate decreases, red upward arrows increases, and equality signs unchanged rates. The error-family percentages are reported separately from the modality-level error rates.

Tab. 10 Modality-wise feedback errors. Error rates (%) on the fixed feedback set; N: group size. Arrows compare consecutive rounds. Bold/underline: two lowest rates; blue: R3.

Modality	N	R0	R1	R2	R3
Radiology	1509	37.18	35.98 ↓	**31.28** ↓	32.21 ↑
Nuclear medicine	234	34.62	26.50 ↓	24.79 ↓	**22.65** ↓
Ultrasound	351	44.73	39.03 ↓	39.60 ↑	**36.18** ↓
Endoscopy	247	32.39	30.36 ↓	27.13 ↓	**21.86** ↓
Pathology	378	34.13	34.66 ↑	30.95 ↓	**28.31** ↓
Ophthalmology	135	34.07	27.41 ↓	22.96 ↓	**20.00** ↓
Dermatology	593	30.35	27.82 ↓	24.62 ↓	**23.10** ↓
Dental imaging	308	40.26	37.01 ↓	37.66 ↑	**32.79** ↓
Functional tests	1245	20.00	17.75 ↓	16.22 ↓	**15.50** ↓

Tab. 11 Error-family feedback profiles. Percentages sum the five subtypes in each family. These aggregates are not directly comparable to modality error rates. Notation follows Tab. 10.

Error family	R0	R1	R2	R3
Visual misreading	14.10	13.79 ↓	11.40 ↓	**10.23** ↓
Reasoning error	12.50	12.20 ↓	10.40 ↓	**9.40** ↓
Differential error	5.40	5.25 ↓	4.20 ↓	**3.90** ↓
Knowledge gap	2.19	2.49 ↑	2.16 ↓	**1.96** ↓

B.4 Detailed Error Statistics for the Current Rounds

Tab. 12 Error-subtype rates (%). Visual/reasoning subtypes are on the left; differential/knowledge subtypes on the right. R1–R5 are reasoning-error codes. Arrows compare consecutive rounds: blue ↓/red ↑/gray = indicate lower/higher/unchanged rates. Bold/underline: two lowest distinct rates per row; blue shading: R3.

Subtype	R0	R1	R2	R3
V1 Missed key finding	3.12	2.95 ↓	2.36 ↓	**2.12** ↓
V2 Spurious finding	2.87	2.82 ↓	2.28 ↓	**2.05** ↓
V3 Location / laterality	2.65	2.68 ↑	2.20 ↓	**1.98** ↓
V4 Sequence / phase	2.91	2.74 ↓	2.32 ↓	**2.08** ↓
V5 Cross-image alignment	2.55	2.60 ↑	2.24 ↓	**2.00** ↓
R1 Omitted evidence	2.88	2.55 ↓	2.16 ↓	**1.95** ↓
R2 Conflicting evidence	2.41	2.42 ↑	2.08 ↓	**1.88** ↓
R3 Unsupported inference	2.67	2.48 ↓	2.12 ↓	**1.92** ↓
R4 Temporal / numeric	2.22	2.35 ↑	2.04 ↓	**1.85** ↓
R5 Answer mapping	2.32	2.40 ↑	2.00 ↓	**1.80** ↓

Subtype	R0	R1	R2	R3
D1 Missing candidate	1.21	1.05 ↓	0.92 ↓	**0.86** ↓
D2 No discriminating cue	1.08	1.12 ↑	0.88 ↓	**0.82** ↓
D3 Premature closure	1.14	0.98 ↓	0.84 ↓	**0.78** ↓
D4 Candidate ranking	0.97	1.08 ↑	0.80 ↓	**0.74** ↓
D5 Exclusion misuse	1.00	1.02 ↑	0.76 ↓	**0.70** ↓
K1 Sign interpretation	0.51	0.48 ↓	0.48 =	**0.44** ↓
K2 Diagnostic criteria	0.44	0.52 ↑	0.44 ↓	**0.40** ↓
K3 Anatomy / mechanism	0.42	0.45 ↑	0.40 ↓	**0.36** ↓
K4 Conditions / exceptions	0.40	0.50 ↑	0.42 ↓	**0.38** ↓
K5 Test / treatment rule	0.42	0.54 ↑	0.42 ↓	**0.38** ↓

Tab. 13 separates repair volume from success rate. Counts measure attempted and successful repairs within each error family.

Reasoning-error repair success rises from 56.43% at R1 to 83.18% at R3, while knowledge-gap repair success falls from 90.28% to 48.82%. These rates describe the repair candidates submitted in each round.

Tab. 13 Repair outcomes by error family. Attempted and successful question counts, and repair success rates (%). Gray arrows track counts; blue/red arrows mark rising/falling success rates. Bold/underline: best/second-best rates per family; blue shading: R3.

Error family	R1			R2			R3		
	Attempted	Successful	Success (%)	Attempted	Successful	Success (%)	Attempted	Successful	Success (%)
Visual misreading	12,189	9,826	**80.61**	10,245 ↓	6,789 ↓	66.27 ↓	10,478 ↑	7,094 ↑	67.70 ↑
Reasoning error	9,355	5,279	56.43	7,857 ↓	6,246 ↑	79.50 ↑	6,789 ↓	5,647 ↓	**83.18** ↑
Differential error	4,064	2,995	**73.70**	6,742 ↑	4,673 ↑	69.31 ↓	4,786 ↓	3,291 ↓	68.76 ↓
Knowledge gap	1,625	1,467	**90.28**	987 ↓	550 ↓	55.72 ↓	1,567 ↑	765 ↑	48.82 ↓

C Medical Taxonomy and Error Categories

C.1 Taxonomy, Permissions, and Execution Contracts

C.1.1 Four-Dimensional Medical Tag Dictionary

The medical taxonomy describes each question along four independent axes: modality, anatomy, disease, and difficulty. Each axis allows multiple hierarchical labels for capability analysis and question retrieval.

Modalities. Computed tomography (CT), magnetic resonance imaging (MRI), positron emission tomography (PET), and single-photon emission computed tomography (SPECT). Other abbreviations include optical coherence tomography (OCT), fundus fluorescein angiography (FFA), indocyanine green angiography (ICGA), cone-beam computed tomography (CBCT), electroencephalogram (EEG), electromyogram (EMG), and electrocardiogram (ECG).

Parent	Second-level scope
1 Radiography	X-ray; CT; MRI; mammography; fluoroscopy; angiography
2 Nuclear medicine	PET/CT; SPECT; radionuclide scans
3 Ultrasound	B-mode; echocardiography; vascular; interventional
4 Endoscopy	Gastrointestinal; bronchial; laryngeal/nasal; bladder; joint; laparoscopic
5 Pathology	Histology; cytology; molecular pathology
6 Ophthalmology	Fundus photography; OCT; FFA/ICGA; slit lamp
7 Dermatology	Dermoscopy; clinical photography; reflectance confocal microscopy
8 Dental	Panoramic; CBCT; periapical; cephalometric
9 Functional tests	ECG; EEG; EMG

Anatomy hierarchy.

Parent	Second-level scope
1 Nervous system	Brain; spinal cord; peripheral nerves
2 Head and neck	Orbit; ear; sinuses; oral/maxillofacial; pharynx/larynx; neck soft tissue
3 Thorax	Lung/airway; mediastinum; pleura; heart; great vessels; breast; chest wall
4 Abdomen	Hepatobiliary; pancreas; spleen; gastrointestinal; renal/urinary; retroperitoneum
5 Pelvis	Male reproductive; female reproductive; pelvic floor; pelvic bones
6 Spine	Cervical; thoracic; lumbar; sacrococcygeal; disc; canal; paraspinal tissue
7 Musculoskeletal	Craniofacial bones; upper limb; lower limb; muscles; periarticular soft tissue
8 Vascular	Arteries; veins; lymphatic system
9 Skin and soft tissue	Head/face; trunk; limbs; subcutaneous; deep soft tissue

Disease hierarchy.

Parent	Second-level scope
1 Neoplasm	Malignant; benign; metastatic; precancerous categories
2 Infection and inflammation	Bacterial; tuberculosis; viral; fungal; parasitic; noninfectious inflammation
3 Vascular disease	Ischemic; hemorrhagic; aneurysm; thrombosis/embolism; malformations; atherosclerosis/stenosis
4 Degenerative disease	Joint; neural; organ degeneration
5 Trauma	Fracture; joint injury; visceral injury; head injury; foreign body
6 Congenital/developmental	Cardiovascular; nervous; skeletal; digestive; urinary; respiratory
7 Metabolic/endocrine	Diabetes complications; thyroid; adrenal; gout; metabolic bone
8 Autoimmune	Lupus; ankylosing spondylitis; multiple sclerosis; neuromyelitis optica spectrum disorder (NMOSD); autoimmune liver; immunoglobulin G4 (IgG4); vasculitis; systemic sclerosis; myositis; Sjogren
9 Functional disorders	Obstruction; effusion; ectopic gas; torsion; stones
10 Iatrogenic/postoperative	Normal postoperative change; complications; implants; post-radiotherapy; drug-related

Leaf specificity and multi-label assignment. CT branches into noncontrast, contrast-enhanced, CT angiography (CTA), high-resolution CT (HRCT), and low-dose CT; MRI includes T1- and T2-weighted imaging, fluid-attenuated inversion recovery (FLAIR), diffusion-weighted imaging (DWI), magnetic resonance angiography (MRA), functional MRI (fMRI), magnetic resonance spectroscopy (MRS), and susceptibility-weighted imaging (SWI). Anatomy can descend from thorax to lung/airway and then to a lung lobe, or from abdomen to pancreas and then head/body/tail. Disease labels descend from a broad family to a mechanism or disease group and then to a named entry. Preserve image IDs for mixed modalities and record every supported anatomical site. **Assign the deepest supported node**, not a guessed leaf; use unknown, not-applicable, or a supported parent when necessary. **Expand ancestors as a set**

and count each question once per group. Related concepts on different paths require an alias ledger, not automatic merging. Log label text, evidence/image ID, confidence, visibility, and taxonomy version.

Difficulty rubric.

Level	Task requirement
L1	Single modality and conspicuous canonical findings; direct recognition.
L2	Common presentation requiring localization or characterization and limited context.
L3	Atypical or multiple findings requiring sequence integration or differential comparison.
L4	Subtle findings, multisystem involvement, or complex context requiring extensive integration.
L5	Highly complex or disputed tasks requiring multidisciplinary, quantitative, or specialized knowledge.

Assign difficulty from the actual question: visibility, number of required comparisons, evidence integration, and specialized knowledge. A disease name, modality, or a single failed rollout does not determine the level. Rare disease alone does not force L5. Record the rationale and adjudicate ambiguous adjacent levels.

C.1.2 Error Codebook: Four Parents and Twenty Subtypes

We specify five subtypes under each of the four existing parent categories. Each row defines an observable diagnostic criterion and a navigation operation; it does not supply a gold answer.

V: Visual misreading.

Code / subtype	Diagnostic criterion	Navigation operation
V1 Missed finding	A supported salient finding is not observed or described.	**Reinspect** the relevant region.
V2 Hallucinated or misidentified finding	An artifact, normal structure, or unsupported feature is read as a lesion.	**Check** direct visual support.
V3 Anatomy or laterality error	The organ, level, compartment, or side conflicts with visible evidence.	**Recheck** landmarks and orientation.
V4 Modality, sequence, or phase error	Acquisition type, sequence, or phase is misidentified, distorting feature reading.	**Verify** acquisition context before reading features.
V5 Cross-image correspondence error	A lesion is matched incorrectly across images, slices, or views.	**Align** images and anatomical locations.

R: Reasoning error.

Code / subtype	Diagnostic criterion	Navigation operation
R1 Omitted evidence	An observed finding or stated fact is omitted from inference.	**Reintroduce** the omitted evidence.
R2 Conflicting evidence integration	Correctly extracted evidence conflicts and is not reconciled.	**Reconcile** supporting and opposing evidence.
R3 Unsupported inference or causal leap	A conclusion lacks the required logical or causal link.	**Restore** a testable intermediate inference.
R4 Temporal or quantitative error	Correctly read facts are combined with an incorrect temporal or numerical relation.	**Check** chronology, units, or calculation.
R5 Task constraint or answer-mapping error	The reasoning violates the requested target, negation, or option mapping.	**Recheck** the requested target and answer mapping.

D: Wrong differential.

Code / subtype	Diagnostic criterion	Navigation operation
D1 Omitted candidate	A necessary evidence-compatible alternative is not considered.	**Broaden** the candidate comparison.
D2 Missing discriminative comparison	Alternatives are listed without comparing discriminative evidence.	**Identify** discriminative evidence.
D3 Premature closure or anchoring	An initial diagnosis is fixed before counterevidence is considered.	**Revisit** counterevidence and alternatives.
D4 Incorrect candidate ranking	Known support, counterevidence, or priors are weighted incorrectly.	**Compare** the strength of evidence.
D5 Misapplied exclusion	An alternative is excluded using insufficient negative evidence or a nonessential criterion.	**Check** whether exclusion is justified.

K: Knowledge gap.

Code / subtype	Diagnostic criterion	Navigation operation
K1 Missing finding interpretation	The required medical meaning of a finding or term is missing.	**Cue** the relevant conceptual relation.
K2 Missing diagnostic criterion	A required diagnostic, grading, or staging rule is unavailable.	**Cue** the relevant rule category.
K3 Missing anatomy or pathophysiology	Knowledge linking anatomy, mechanism, and manifestations is missing.	**Bridge** the minimal mechanistic relation.
K4 Missing applicability condition or exception	A general rule is known but its prerequisites or exceptions are missing.	**Cue** the applicability check.
K5 Missing investigation or management knowledge	A next-investigation or management task requires an unavailable principle.	**Cue** the decision principle without the answer.

Boundary rules and reporting. A finding not perceived is V1; a finding already perceived but omitted from inference is R1. Incorrect correspondence across images is V5; conflicting correctly extracted evidence is R2. Candidate-set construction, comparison, and exclusion belong to D; a demonstrated missing prerequisite rule belongs to K. Knowledge-gap labels require evidence of a missing prerequisite rule. **Select the earliest consequential supported error** as the primary subtype and record later failures as secondary labels, with step spans and evidence. If the parent is clear but the subtype is not, retain the parent and an unresolved-subtype status; if even the parent is unsupported, mark the audit unresolved. These are annotation states, not extra error classes. Use **one primary subtype per trajectory** for additive counts; report secondary labels separately. Aggregate V1–V5 into V, and similarly for R, D, and K. Re-audit fresh rollouts at each checkpoint using subtype-specific annotations and denominators.

C.1.3 Information Access Matrix

Component	Permitted inputs	Excluded inputs
Feedback evaluator	Feedback items, labels, current predictions; emits aggregates and transitions	Final-test results for planning; feedback export into SFT
Planner / retrieval	Aggregate feedback, training tag metadata (including true disease labels), coverage, budgets	Final-test items/results; any authority to forward answer-derived labels to repair
Auditor / hint renderer	Original images, question/options, current failed trajectory, entropy profile, observable tags with provenance	Gold answer, answer-derived disease/location, reference rationale, hidden retrieval labels
Continuation M_t	Original input and assistant prefill $p_b \oplus g(h)$	Gold answer, full audit, hidden tags, teacher solution
Admission / exporter	Candidate, **reference answer** for verification, boundary, provenance, suffix scores	Feedback/test examples as training targets
SFT trainer	Frozen training manifest; $p_b \oplus g(h) \oplus s^+$, masks	Full audit as model input; feedback/test samples

Use **explicit allowlist serialization**, not deletion of one answer field from an otherwise unrestricted object. **Reject fields without observable provenance**, including filenames/IDs encoding labels. An authorized evaluator can identify a wrong training answer before repair without transmitting the answer. Generation sees only a boolean routing decision, not the key. Record hashes of redacted requests for leakage audit. The planner's chosen hint family is passed through an eligibility check derived from observable context and current diagnosis; disease-bearing planner explanations are never forwarded.

C.1.4 Structured Fields and Plan Validation

Use JavaScript Object Notation (JSON). Missing measurements are null with reasons; zero is measured zero.

Question metadata: canonical question ID; source ID; patient/case/article cluster IDs when available; image hashes and near-duplicate cluster; split; taxonomy version; per-image modality; all four tag sets; per-tag provenance, visibility (statistical or observable), and uncertainty.

Feedback record: checkpoint hash; split hash; group and resolved parent; unique sample count; error count/rate; all four transition counts; both regression denominators; current error-type/subtype counts; confidence interval; training-pool unique availability; measured repair/admission/cost fields.

Plan: round ID; generator and initialization hashes; planner type/version and seed; feedback hash; target group IDs; numeric feedback evidence and fallback ancestry; training sources; selected IDs; attempt/call/token ceilings;

per-group quota; permissible hint operation; history replay allocation; replacement rule; supervised-token cap; shortfall/reallocation policy; **fixed training-config hash**; acceptance-validator result.

Repair record: question/checkpoint IDs; rollout hash; step/token spans; entropy profile/proposal; error type/subtype; audit boundary; verbatim prefix span; observable-tag payload hash; recovery cue; continuations/seeds; correctness/integrity checks; student suffix scores and cutoffs; reason for admission or rejection; all tokens generated or scored.

Mix and training record: ordered example IDs; source round and generator; dedup/replacement ledger; historical/new role; token masks; sequence lengths; unit weights; packing map; effective repetitions and exposures; optimizer and scheduler reset flags; learning rate/warmup; initialization hash; output hash; second-stage and cumulative costs.

Execution receipt: planned and actual quotas, unique selected and attempted failures, successful and admitted repairs, substitutions, exhausted budgets, rejected records, uncovered groups, and actual training tokens.

Reject cross-split IDs, duplicate quota charges, exceeded budgets, out-of-pool retrieval, unsupported hints, missing visibility provenance, and changed training hyperparameters. Save plans before generation; log counts.
The main-agent feedback and next-round planning prompt is given in Appendix D.2; the trajectory audit prompt is given in Appendix D.3.

C.1.5 Training Configuration and Evaluation Scope

Training uses brain floating-point 16 (BF16) precision on graphics processing units (GPUs). Parallelism abbreviations are tensor parallelism (TP), expert parallelism (EP), pipeline parallelism (PP), context parallelism (CP), expert tensor parallelism (ETP), and data parallelism (DP); the following number gives the parallelism degree. Configuration keys retain their literal software spelling: llm denotes large language model, vit denotes vision transformer, cpu denotes central processing unit, moe denotes mixture of experts (MoE), gemm denotes general matrix multiplication, mlp denotes multilayer perceptron, and lr denotes learning rate.

Tab. 14 MoE training configuration. Recorded settings for model training.

Configuration	Value	Configuration	Value
Framework	ms-swift + Megatron-LM backend	add_non_thinking_prefix	true
Precision	BF16	loss_scale	ignore_empty_think
micro_batch_size	1	MAX_PIXELS	3145728
global_batch_size	64	dataset_num_proc	96
packing	true	dataloader_num_workers	8
attention_backend	flash	logging_steps	1
calculate_per_token_loss	true	eval_iters	1000
freeze_llm /freeze_vit / freeze_aligner	false /true /true	recompute_granularity	selective
optimizer_cpu_offload	true	sequence_parallel	true
use_precision_aware_ optimizer	true	max_length	28672
offload_fraction	1.0	epochs	2
moe_grouped_gemm	true		
moe_shared_expert_overlap	true	save_total_limit	2
moe_aux_loss_coeff	1e-6	Recompute modules	core_attn + moe + mlp
lr	1e-5	moe_permute_fusion	true
lr_warmup_fraction	0.05	Parallelism	TP4/EP8/PP1/CP1/ETP1; 64 GPUs
min_lr	1e-7		

EP and MoE-specific options apply only to MoE models.
Both example types use full-response loss, retaining recovery cues in repairs.
§ 4 reports accuracy on seven benchmarks and their mean, comparing each backbone's best complete checkpoint with Base.

D Prompts

D.1 Shared Question-Answering Prompt

All question-answering tasks use the following instruction verbatim.

Shared Question-Answering Instruction

Please reason step by step, and put your final answer within `\boxed{}`.

D.2 Main-Agent Feedback and Next-Round Planning

The following template connects feedback diagnosis to next-round selection and repair priorities under the method protocol. It specifies agent behavior; run-specific inputs and execution records are required to document its use.

Main Agent: Feedback Summary and Next-Round Data Plan

You are the feedback and data-planning agent for MedEvo. Diagnose model weaknesses and produce the next data plan. Do not write solutions to training questions or change the experimental comparison settings.
Inputs. Round and checkpoint; fixed 5,000-question feedback results for current and previous checkpoints; four-axis medical tags; four error families and their twenty subtypes; per-group sample counts, correctness transitions, and current error diagnoses; available training-question IDs and tags; historical trajectory records; fixed generation/repair budgets, support threshold, regression weight α, and admission rules. Report missing inputs.
1. Summarize feedback. For each supported tag group, report error rate, round-to-round change, corrected errors, regressions, and persistent errors. Summarize four-family and twenty-subtype distributions with explicit denominators. Count each question once within a group; do not sum overlapping groups into an overall rate. Merge unsupported groups into supported parents. Distinguish observations from unverified explanations.
2. Set priorities. Rank groups using $S_{g,t} = \epsilon_{g,t} + \alpha N^{10}_{g,t}/n_g$ under the supplied configuration. Identify dominant or persistent error types within each priority group. State which measured feedback supports each priority; do not infer that a high-error modality contains every error subtype.
3. Select training questions. Allocate the fixed budget proportionally to group scores, cap quotas by available unique questions, and redistribute unused quota. Preserve modality coverage and apply the configured fallback when all scores are zero or support is insufficient. Retrieve by medical tags, using historical errors only as prioritization cues. Deduplicate questions; return training IDs, quotas, and shortfalls. **Exclude feedback/test examples.**
4. Guide fresh rollouts and repair. Ask the current checkpoint to generate the configured eight rollouts per selected question. **Diagnose their actual errors again.** Use feedback-identified modality–error pairs to prioritize failed training trajectories: prefer joint matches, followed by modality-only or error-type-only matches when necessary. For each selected trajectory, pass its observable medical tags and current error diagnosis to the hint generator. The recovery hint must address the error actually observed in this trajectory. Send observable tags and local diagnoses to the auditor and hint generator, **without reference answers or answer-bearing metadata**. Retain correct rollouts and accepted continuations under the existing screening rules; if repair fails, use an available quality-screened correct rollout. If none is available, retain a valid historical target or report none accepted.
5. Plan the next round. **Keep at most one active target per question** and replace it only with an accepted improvement. **Train on the complete updated dataset from M_0.** Keep training settings and evaluation protocol fixed. Evaluate the next checkpoint on the same feedback set to measure the targeted changes; independent benchmark results are reported separately and must not drive selection.
Output. Return a structured plan containing: round and checkpoint; feedback summary and denominators; priority tag/error pairs and supporting statistics; score and quota per group; selected unique training IDs; rollout and repair budgets; repair priorities; dataset-retention/replacement rules; initialization M_0; next-round feedback comparisons; and missing inputs or budget shortfalls. After execution, record actual generated, repaired, admitted, and replaced counts.

D.3 Stage 1 Audit Prompt

The diagnostic model receives the image, the question with options, the failed trajectory, the complete step-level uncertainty profile, and the entropy-derived anchor and candidate window. The auditor is gold-blind: **the reference answer is not an input to diagnosis**. Answer-derived disease labels, hidden lesion positions, reference explanations, and planner retrieval metadata are also excluded. Only provenance-checked observable tags are allowed. The entropy proposal is a localization prior, not an error label: the auditor inspects the proposed region first, traces its evidence dependencies backward, and may move the final boundary when the medical or logical evidence requires it. It returns a structured audit that identifies a compact revision region and constrains the subsequent continuation.

The emitted hint must remain answer-blind. Fill brace-delimited placeholders at runtime.

Prompt D.3: Stage-1 Entropy-Guided Medical Trajectory Audit

You are a senior medical reviewer auditing a failed multimodal reasoning trajectory segmented into semantic steps $s_1, \ldots, s_K$.

You are given:

(1) the medical image(s);

(2) the clinical question and answer options;

(3) the failed student trajectory segmented into semantic steps;

(4) STEP_ENTROPY: the uncertainty score E_k^* of every step;

(5) NORMALIZED_ENTROPY: the normalized uncertainty score of every step;

(6) LOCAL_SURGE: the local increase ΔE_k relative to the preceding step history;

(7) ENTROPY_ANCHOR: the step proposed by the uncertainty locator;

(8) ENTROPY_WINDOW: the compact candidate window surrounding that anchor, and ENTROPY_THRESHOLD if a threshold was used.

(9) OBSERVABLE_TAGS: per-image modality and other context supported by the original input, with evidence provenance; never hidden answer-derived tags.

The entropy signal guides audit localization. High uncertainty can reflect a genuine failure, but it can also arise from difficult terminology, ambiguous visual evidence, or a medically valid differential diagnosis. Combine the entropy proposal with the image, clinical evidence, and dependencies between reasoning steps to identify the first consequential failure.

Error-conditioned hint construction. Construct the hint from OBSERVABLE_TAGS, ERROR_TYPE, ERROR_SUBTYPE, and KEY_ERROR. Select a recovery operation addressing the diagnosed failure:

- **Visual misreading:** direct attention to the relevant region, visual attribute, or cross-image correspondence.
- **Reasoning error:** revisit an unsupported inference or reconnect observed evidence to the requested decision.
- **Differential diagnosis error:** compare discriminative evidence or reconsider an underexamined alternative.
- **Knowledge gap:** identify the relevant concept, rule, or applicability condition to reconsider.

Specify one concrete next operation grounded in the current question and trajectory. Express it as a brief recovery cue without supplying the corrected finding, final diagnosis, or answer option.

Audit procedure:

(1) Inspect ENTROPY_WINDOW first. Determine whether its local entropy peak or surge corresponds to an unsupported observation, invalid inference, incorrect differential comparison, or missing medical principle.

(2) Trace evidence dependencies backward from ENTROPY_ANCHOR. Decide whether the decisive failure begins inside the proposed window or an earlier step introduced the unsupported premise that caused the later entropy increase.

(3) Do not automatically select the highest-entropy step. Justify the final AUDIT_ANCHOR and SAFE_PREFIX_END using the medical image, clinical information, and logical dependencies.

(4) Prefer a compact localized repair and preserve every preceding step that remains medically and logically valid.

(5) You may move the audit anchor outside ENTROPY_WINDOW when the proposal misses the first consequential failure, but briefly explain why the entropy signal was misleading.

Return a JavaScript Object Notation (JSON) object with the following fields, grounded strictly in the provided image and text:

(a) ACTION: REPAIR or SKIP.

(b) STEP_SUMMARY: a brief one-sentence semantic summary for each step.

(c) ENTROPY_ASSESSMENT, containing: proposed_anchor, proposed_window, proposal_status, and brief_reason. proposal_status must be one of {HIT, PARTIAL_HIT, MISS, NON_ERROR_UNCERTAINTY}. HIT means that the first consequential failure is at or immediately adjacent to the proposed anchor; PARTIAL_HIT means that the proposal covers the propagated error region but the unsupported premise begins earlier; MISS means that the first failure lies outside the proposed window; and NON_ERROR_UNCERTAINTY indicates a medically or logically valid high-entropy region.

(d) AUDIT_ANCHOR: the index of the first consequentially incorrect step.
(e) SAFE_PREFIX_END, containing step_index and the exact verbatim_ending_span of the last fully valid step before the repair region.
(f) AUDIT_WINDOW: a short contiguous sequence of steps that should be reconsidered.
(g) ERROR_TYPE and ERROR_SUBTYPE: checkpoint-specific diagnosis, plus KEY_ERROR: the localized error stated without revealing the correct answer.
(h) HINT_TYPE: one of {perceptual_reexamination, inference_reconstruction, differential_disambiguation, knowledge_bridging}.
(i) MINIMAL_HINT: a brief recovery cue with flexible wording, conditioned on observable medical tags and the diagnosed local error. Name the evidence or reasoning operation to revisit. Do not include the error-category label or audit explanation. Introduce the cue with a natural transition; no fixed phrase or first-person form is required.
(j) IMAGE_CHECK, containing applicable (true or false) and target (the specific modality, anatomy, laterality, finding, relation, or measurement to re-inspect).
(k) KNOWLEDGE_SCOPE: the narrow medical principle to recall, or null.
(l) DIFFERENTIAL_FOCUS: the discriminative axis to compare, or null.
(m) STOP_CONDITION: the evidence condition under which the continuation should commit to an answer and stop.
Constraints:
- Do not access hidden disease labels or hidden lesion coordinates.
- Do not state a gold answer, corrected diagnosis, or correct option letter.
- Do not provide a replacement chain of thought or teacher-authored solution.
- Do not introduce symptoms, history, tests, measurements, or visual findings absent from the input.
- Do not discard a valid step merely because its entropy is high, and do not preserve an invalid step solely for low entropy.
- SAFE_PREFIX_END must precede the first consequential failure.
- AUDIT_WINDOW should normally overlap ENTROPY_WINDOW, but may deviate when supported by explicit medical or logical evidence.
- If only final-answer formatting or option selection is wrong and no reasoning revision is justified, set ACTION to SKIP.

D.4 Stage 2 Continuation Prompt

The continuation input contains the original question and an unfinished assistant message with the retained prefix and an error-conditioned recovery cue. Stage 1 constructs the cue from observable medical tags, ERROR_TYPE, ERROR_SUBTYPE, and KEY_ERROR. Feedback determines repair priority; the selected trajectory's own diagnosis determines cue content. The same student continues from this prefilled message.

Prompt D.4: Stage-2 Assistant-Prefill Continuation

User message:
{original medical image(s), question, and answer options}

Assistant message:
{safe_prefix}
{Natural transition} {MINIMAL_HINT from Stage 1}

MINIMAL_HINT is inserted verbatim after the retained prefix. The student receives only the cue; error labels and audits remain metadata. Continue immediately after the cue in the same assistant message; preserve the prefix. The SFT export retains the student-style recovery cue. Repaired and original correct examples use full-response loss in Equation 13, supervising all assistant-response tokens. No audit or external hint is supplied at final evaluation. Record exported examples and token masks with the training configuration.

E Case Study

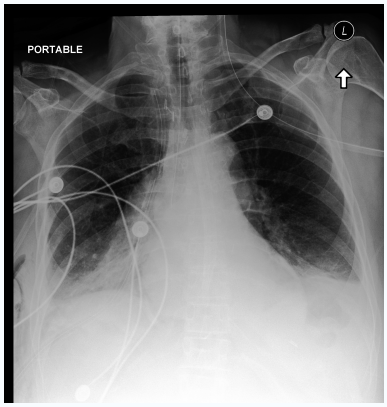
We present two medical image question-answering cases, including the question, options, and excerpts from the student's initial reading and subsequent continuation, with the hint marked separately. We preserve the wording of the student outputs without language editing. Purple labels separate the original text, hint, and continuation; blue emphasis marks selected observations and reasoning actions in the quoted outputs.

E.1 Case 08: Chest radiograph

The initial reading is shown as a failed-rollout excerpt for comparison; it is not the safe prefix retained after rollback. The hint and continuation show how the initial interpretation is revised. Clinical terms: anteroposterior (AP), electrocardiogram (ECG),
nasogastric (NG), chest radiograph (CXR), pulmonary embolism (PE), tuberculosis (TB).

Original question and options

A 67-year-old man is recovering in the surgical intensive care unit following an esophagectomy. A portable anteroposterior chest radiograph (Figure A) is obtained to evaluate his postoperative status. The patient has a right-sided chest tube in place. Based on the radiographic findings, which of the following is the most likely cause of the bilateral lower lung opacities? Options:
A. Pleural effusion
B. Pulmonary embolism
C. Emphysema
D. Cavitary tuberculosis
E. Tension pneumothorax
Please reason step by step, and put your final answer within \boxed{}.



Student model output

Initial reading from the failed rollout (before rollback)

<think>

The user wants a step-by-step reasoning and a final boxed answer. Let me look at the radiograph carefully. It's a **portable AP chest radiograph** of a 67-year-old man post-esophagectomy. There's a right-sided chest tube. There are ECG leads overlying the chest. There's an NG tube or feeding tube coursing down the midline (post-esophagectomy, possibly gastric pull-up). The question asks about **bilateral lower lung opacities**. Looking at the image, the lower zones do not show convincing increased opacity; the costophrenic angles are not clearly blunted, and there is no definite meniscus-like contour. The hemidiaphragms remain relatively well defined. So pleural effusion is not strongly supported.

Navigation hint

Before weighing the choices, **re-check the morphology of the basal opacities**: **costophrenic-angle blunting, a meniscus-like contour**, and partial obscuration of the hemidiaphragms would localize the process primarily to the...

Student continuation after the hint

Now, **let me think about the other options**: pulmonary embolism doesn't usually cause bilateral lower lobe opacities (may cause atelectasis or infarction, but not typical). Emphysema causes hyperlucency, not opacity. Cavitary TB causes upper lobe cavities, not bilateral lower lobe opacities. Tension pneumothorax causes hyperlucency and mediastinal shift, not opacity. So the answer is A, pleural effusion.

The question says "bilateral lower lung opacities" — in
a postoperative patient with a chest tube, bilateral pleural effusions (or a chylothorax after esophagectomy) are common. The radiographic appearance of basal opacities with meniscus and blunting of the costophrenic angles supports pleural effusion.

Therefore the

Let me describe the radiographic findings in more detail: bilateral lower zone opacities with a meniscus sign, blunting of the costophrenic angles, obscuration of the hemidiaphragms; the chest tube is on the right; the left side shows a similar opacity. No evidence of hyperlucency or pneumothorax; no cavitary lesions; no findings of emphysema (which would be lucent). PE is not a radiographic diagnosis on chest X-ray anyway and wouldn't present as bilateral basal opacities.

Therefore the answer is A.

I should present step-by-step reasoning: identify the findings, exclude the options, and conclude with A.

Let me double-check
the image description once more: the left costophrenic angle is blunted with a meniscus; the right base shows opacity consistent with effusion and possibly some atelectasis. The question asks for the "most likely cause" — pleural effusion.
Also, after esophagectomy,
patients often develop pleural effusions due to chylothorax or transudate from fluid shifts. So A is clinically plausible as well.
Format: the answer should be moderately detailed, with step-by-step reasoning, and a final answer. Use headings or bullet points sparingly. Let me write a concise structured analysis: findings, differential exclusion, conclusion. Then give the final
Let me write it like this:
Step 1 — Systematic reading: portable AP film; cardiac
silhouette is difficult to assess; bilateral lower zone opacities; blunting of the costophrenic angles with a meniscus-like upper border; obscuration of the hemidiaphragms; right-sided chest tube (likely for drainage of pleural fluid/air); no evidence of pneumothorax (no visceral pleural line with hyperlucency); no cavitary lesions; lung volumes are not hyperinflated.
Step 2 — Match with options:
- Pleural effusion: fits — dependent (basal) opacities with meniscus, blunting of the costophrenic angles, common after esophagectomy (including chylothorax).
- PE: not visible on CXR; wouldn't cause diffuse basal opacities.
- Emphysema: causes hyperlucency, flattened diaphragms — opposite of opacity.
- Cavitary TB: upper lobe cavitary lesions, not bilateral basal opacities.
- Tension pneumothorax: hyperlucent hemithorax, mediastinal shift away — not present.
Conclusion: A.
I might
also mention that the chest tube on the right is for drainage of pleural fluid/air, and that the left effusion is likely also present.
Let me double-check the arrow in the image: it points to the left shoulder/clavicle area — probably just a marker. Not relevant.
Final
I'll write the response
with clear reasoning. Format: short paragraphs or bullet points. Let me use a structured but concise approach</think>
Most appropriate answer: \boxed{A}.

E.2 Case 31: Elbow radiograph

Clinical abbreviations in this case: anteroposterior (AP) and fall on an outstretched hand (FOOSH).

Original question and options

A 32-year-old patient presents to the emergency department with acute right elbow pain, swelling, and limited range of motion following a fall onto an outstretched hand. An anteroposterior radiograph of the right elbow is obtained (Figure A). What is the most likely diagnosis? Options:
A. Normal elbow radiograph
B. Isolated elbow dislocation
C. Elbow fracture
D. Advanced osteoarthritis
E. Osteochondritis dissecans
Please reason step by step, and put your final answer within \boxed{}.



Student model output

Original context and initial reading
<think>
The user wants a step-by-step reasoning for a radiograph
interpretation question. **Let me carefully look at the image**. The image is labeled "R" (right) and shows an elbow radiograph. The question states it's an AP radiograph, though the image looks somewhat like an oblique or lateral-ish view, but
let's focus on the pathology. Looking at the radiograph: I see the **distal humerus, the proximal radius, and proximal ulna**.

Navigation hint
Before committing,
trace the radial head and neck cortices to verify that the horizontal lucency causes **true cortical interruption or fragment displacement** rather than projectional overlap, then **assess whether the radiocapitellar and ulnohumeral relationships** are...

Student continuation after the hint

The radiocapitellar
alignment appears normal (the radial head is opposite the capitellum), and **the ulnohumeral articulation is also maintained**, suggesting no dislocation. In case of an isolated dislocation, the radial head would be displaced from the capitellum. Here the alignment looks preserved, so an isolated dislocation is unlikely. The presence of a fracture line at the radial head suggests a radial head fracture, so the answer is C (elbow fracture). Osteoarthritis wouldn't present acutely after trauma, and there are no osteophytes. Osteochondritis dissecans is a chronic condition with a loose body, not an acute trauma presentation.

Therefore, the most likely diagnosis is C: elbow fracture (radial head fracture). Let's describe the reasoning steps:

1. Verify patient history: 32-year-old, fall on an outstretched hand
(FOOSH) -> typical mechanism for radial head/neck fractures, coronoid fractures, lateral collateral ligament injuries, etc.

2. Radiographic findings: AP view of the right elbow shows a lucent line with cortical disruption at the radial head/neck, with some displacement/fragmentation - consistent with a radial head fracture. The radiocapitellar and ulnohumeral joints remain congruent, excluding dislocation. No evidence of advanced osteoarthritis (no joint space narrowing, osteophytes, subchondral sclerosis). No chronic loose body or cartilage defect suggestive of osteochondritis dissecans.

3. Therefore the answer is C.

Also, it's worth mentioning that
the "fat pad sign" may be present (elevated anterior fat pad), indicating an intra-articular fracture/hemarthrosis, supporting the diagnosis of a fracture. In the image, there may be a faint anterior fat pad sign. But the key is the radial head fracture.

Let me double-check
the image description once more: the image shows the humerus, ulna, and radius. In the proximal radius region, there's a clear horizontal white line - a fracture of the radial head. The radial head may also be comminuted. So the answer is C.

Let me also consider the possibility of a coronoid fracture. Regardless, it's an elbow fracture.

Thus, the final

Format: The answer should be moderately detailed, including step-by-step reasoning: mechanism, radiographic findings, exclusion of other options, conclusion. Use headings or bullet points. Final

Let me carefully
describe the image: In the image, the radial head shows a transverse lucency consistent with a fracture; the radial head fragment is slightly displaced; the radiocapitellar relationship is preserved; the ulnohumeral angle is normal; no dislocation.

Let me also mention that the lateral fat pad sign supports an intra-articular fracture.

Let me write the response.

Also, the question states "anteroposterior radiograph," but the image is somewhat lateral; I won't overemphasize the view.

Final

Step 1 – Mechanism of injury: A fall onto an outstretched hand (FOOSH) is the classic mechanism for a radial head/neck fracture, coronoid fracture, or lateral collateral ligament injury in a young adult.

Step 2 – Systematic review of the radiograph:

- **Joint alignment:** The radiocapitellar and ulnohumeral relationships are preserved – the radial head remains opposite the capitellum and the ulna articulates normally with the trochlea. This argues **against an isolated elbow dislocation** (option B), which would show obvious displacement of these relationships.

- **Bony detail:** There is a transverse lucent line with cortical disruption and a small displaced fragment at the radial head/neck region – i.e., a **fracture of the radial head**. A faint anterior fat-pad sign (sail sign) is also suggestive of an intra-articular fracture/hemarthrosis.

- **No degenerative changes:** There is no joint-space narrowing, osteophyte formation, or subchondral sclerosis, excluding **advanced osteoarthritis** (option D), which would also be unlikely to present as an acute traumatic event.

- **No chronic lesion:** There is no well-circumscribed subchondral defect with a loose body, which would suggest **osteochondritis dissecans** (option E); that condition is chronic/overuse-related, not acute post-traumatic.

- The film is clearly **not normal** (option A) given the visible fracture line and fragment.

Step 3 – Synthesis: Acute trauma + FOOSH mechanism + visible fracture at the radial head with preserved joint alignment = a fracture of the elbow (radial head fracture), not a dislocation or degenerative/chronic lesion.

Most likely diagnosis: C. Elbow fracture</think>

Most appropriate answer: \boxed{C}.

F Additional Related Work

This section expands the main-text review of medical evaluation, supervision quality, localized correction, and alternative forms of self-evolution. It also covers complementary work in agent safety, robotics, and data generation.

F.1 Medical Reasoning, Distillation, and Data Selection

Medical vision-language models use biomedical instruction tuning [2, 3], generalist reasoning with **Lingshu** [11], and English–Arabic medical image understanding in **BiMediX2** [12]. Verifier-guided search, knowledge-enhanced synthesis, and reinforcement learning[40, 41, 42], together with parallel reasoning and multi-agent collaboration[43, 44, 45], improve reasoning capabilities. Process-level diagnostics and distillation audits distinguish answer correctness from reliable visual recognition, knowledge application, and evidence integration[15, 16, 46].

MMVistaReason [47] combines open-data curation with capacity-aware post-training for multimodal reasoning.

Medical hallucinations expose evaluation limits. In controlled experiments, Farrow et al. [17] show that clinical hallucinations outside predefined criteria can leave rubric scores unchanged. This motivates explicit factuality checks.

Process supervision provides intermediate-step feedback[18], while trajectory synthesis and selection address rationale quality, difficulty, and distribution[48, 49]. **ChainPrune** [19] organizes self-generated reasoning paths into a tree and selects paths that reduce redundancy while preserving essential reasoning steps. Its distinction between token length and structural efficiency motivates content-based trajectory selection. **On-policy distillation** emphasizes compatibility between student states and teacher capabilities[9]; reference-induced shortcuts, reasoning paths, and state coverage also shape supervision value[50, 51, 52]. **MedEvo** connects medical error diagnosis with student-centered supervision, guiding correction with minimal hints while preserving student reasoning.

F.2 Learning from Failed Trajectories and Localized Repair

Failed trajectories contain reusable reasoning as well as errors. Self-Refine and Reflexion improve subsequent attempts through self-feedback and verbal memory without weight updates [53, 54]. **Learning from Mistakes (LEMA)** constructs mistake–correction pairs, **TrajFusion** combines incorrect and correct trajectories, and Exploring Expert Failures (EEF) extracts useful actions from failed demonstrations [6, 10, 55]. Hindsight Hint Distillation (**HHD**) and FailForge guide recovery with concise hints or skills instead of full teacher solutions [1, 56].

Local recovery also depends on where guidance enters and how it becomes supervision. **Error-localization** studies, backtracking-based distillation, Test-Time Scaling via Error Localization (**TTEL**), and Selective Regenerative Decoding investigate localization, backtracking corrections, and prefix reuse [5, 21, 22, 57]. Self-Correction via Reinforcement Learning (**SCoRe**) learns self-correction through reinforcement learning; Project Aletheia and Pyligent derive supervision from verified backtracking traces or search trees; micro-reflective self-distillation combines micro-reflective trajectories with capability-dependent selection [24, 58, 59, 60]. EvoThink and Reflective On-Policy Self-Distillation (ROSD) instead use preference learning and reflective on-policy self-distillation [61, 62].

MedEvo uses medically grounded, **same-student local navigation**: preserving supported prefixes, guiding continuation with gold-blind auditing and tag-conditioned hints, and selecting learnable supervision.

F.3 Data Evolution and Recursive Self-Improvement

STaR and ReST^{EM} select correct self-generated reasoning for supervision[7, 8]. EvoThink and **ROSD** use preference learning and reflective self-distillation[61, 62]; **capability-dependent selection** matches supervision and students [24].

Recursive self-improvement examines how feedback changes subsequent learning experience [63]. In clinical interaction, **MediSkill-Evo** [25] evolves validated clinical, process, symbolic, and visual knowledge banks without fine-tuning the backbone, using process constraints to guide evidence-grounded actions. **MedEvo** instead updates model weights through SFT on repaired student trajectories: it diagnoses medical groups and error patterns on a fixed feedback set, then generates and repairs training-pool trajectories for subsequent **self-distillation**.

F.4 Failure Feedback beyond Medical Reasoning

Failure-aware approaches also organize recovery using structured failure feedback in agent safety and robotics [64, 65, 66].

F.5 Analysis of Frontier-Model Reasoning

Luo et al. [67] study frontier-model reasoning efficiency, step types, and tree structure alongside outcomes.

F.6 Broader Data and Environment Evolution

Difficulty-guided chart synthesis and multimodal generation workflows expand the supervision distribution [68, 69]. Data-strategy optimization, evolving environments, and autonomous medical curricula provide broader settings for recursive improvement [70, 71, 72]. MedEvo evolves data for medical reasoning repair and supervision.

G Additional Baseline Details

Tab. 2 compares external models and trajectory-supervision baselines.

G.1 Backbones and External Comparison Models

We use Qwen3.5-9B, scaling to Qwen3.5-35B and Qwen3.5-397B [28]. A Base result refers to the corresponding initial checkpoint before MedEvo training. Later-round models are trained from that **same initial checkpoint** using the updated supervision dataset; the current checkpoint generates its training trajectories.

The external comparison set includes general-purpose models and models specialized for medicine. Medical models include LLaVA-Med [2], HuatuoGPT-Vision [36], BiMediX2 [12], Lingshu [11], and HealthGPT [39]. InternVL2.5 and InternVL3 provide general-purpose multimodal comparisons [37, 38]. Tab. 2 groups models by proprietary access and open-source parameter scale. These comparisons contextualize performance; they differ from MedEvo's **controlled ablations**.

G.2 Trajectory-Supervision Baselines

Baseline 1: HHD. Hindsight Hint Distillation uses concise hints to scaffold recovery [1]. Its role in the comparison is to assess hint-based supervision alongside MedEvo's **medical diagnosis, prefix preservation, and feedback-driven data updates**. The method follows the cited work; Tab. 2 reports our medical-benchmark evaluation.

Baseline 2: TrajFusion. TrajFusion combines incorrect and correct trajectories to construct training supervision [10]. It provides a comparison for learning from failed reasoning. MedEvo instead constructs a **corrected suffix** from the same student's retained prefix and uses medical feedback to prioritize later repairs.

Baseline 3: Teacher SFT. This baseline uses SFT on supervision generated by Qwen-3.8-Max. It compares **teacher-generated supervision** with MedEvo's repaired student trajectories. This isolates supervision.

G.3 Reading the Comparisons

The medical icon identifies models specialized for medicine, and the repair icon identifies trajectory-repair methods. Blue shading identifies MedEvo. Bold and underlining indicate the highest and second-highest distinct scores within each model group and column, with ties sharing a rank. The **seven-benchmark average** weights each benchmark equally. It is distinct from repair success, the fraction of submitted questions corrected.

Component ablations test full rewriting, suffix continuation, medical-tag feedback, and error-type feedback. Their purpose is to isolate design choices within the reported setup. Training configuration and information-access rules are provided in Appendices C.1.5 and C.1.3, respectively.